



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

29-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,C	A,B	A,C,D	A,B,D	A,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	D	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.72 to 0.75	0.34 to 0.36	37.50	1.25	0.75	0.58 to 0.59	1.73	125.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,D	B,D	B,C	A,B,C	A,C,D	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	A	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.25	0.49	6.00	120.00	15.00	4.00	52.50	3.00

PART-3 : MATHEMATICS

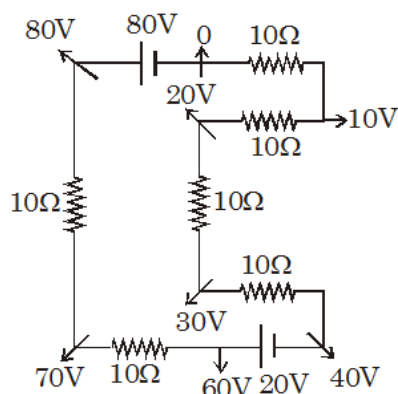
SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,C,D	A,C	A,D	A	A,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	B	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.25	3.00	16.00	8.00	7.00	0.46 to 0.48	3.00	4.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C,D)



$$i = 1A$$

$$V_A - V_B = 40V$$

$$\theta_1 = 30$$

$$\theta_2 = 50$$

2. Ans (A,C)

$$\frac{kq}{r} = \frac{V}{2}k \text{ \& } \frac{kq'}{r} = \frac{-V}{2} \Rightarrow q = 2\pi\epsilon_0 rV$$

$$\text{Total charge flows} = q - q' = \frac{r}{k}V$$

$$d = \frac{1}{2} \times q \times \frac{V}{md}t^2$$

$$\frac{2md^2}{2\pi\epsilon_0 rV^2} = t^2 \Rightarrow t = \frac{\sqrt{md}}{\sqrt{\pi\epsilon_0 rV}}$$

$$i_{avg} = \frac{q}{t} \Rightarrow i_{avg} = \frac{2(\pi\epsilon_0 r)^{3/2}V^2}{\sqrt{md}}$$

Average power loss =

$$P = \frac{\Delta U}{t} = \frac{q(V/2 + V/2)}{t}$$

$$= \frac{qV}{t} = \frac{(2\pi\epsilon_0 rV)V}{\frac{\sqrt{md}}{\sqrt{\pi\epsilon_0 rV}}} \Rightarrow P \propto V^3$$

3. Ans (A,B)

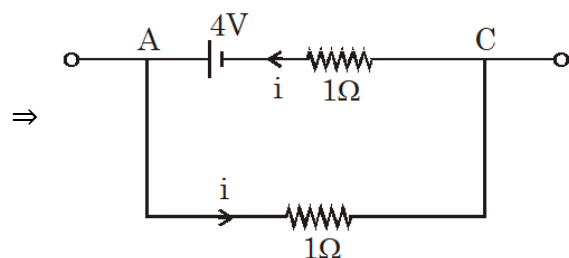
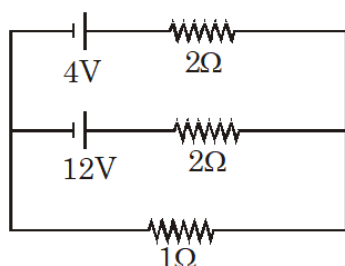
When S_1 is closed, null point cannot be obtained because secondary cell of 4V is in same polarity and p.d. cannot be compared.

When S_7 is closed, null point length

$$= \frac{12\text{V}}{\text{Potential gradient}}$$
$$= \frac{12\text{V}}{8\text{V/m}} = 1.5\text{m}$$

When S_1 and S_2 are both closed, p.d. across terminal = 4V with A at high potential, So null point can be obtained.

When S_1 , S_2 & S_3 are closed



$$\Rightarrow V_A - V_C = 2V$$

$$i = \frac{4}{2} = 2A$$

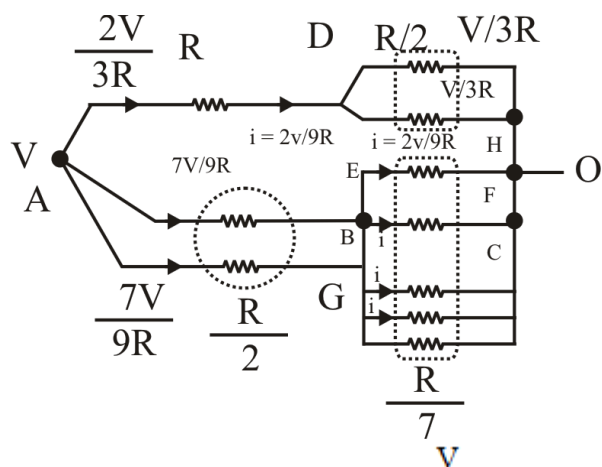
$\therefore$ Balance point obtained at

$$\ell = \frac{2V}{\text{Point gradient}} = \frac{2V}{8V/m} = 0.25m$$

= 25 cm from A.

4. Ans (A,C,D)

After redrawing circuit & taking potential at A = E
& F to be zero.

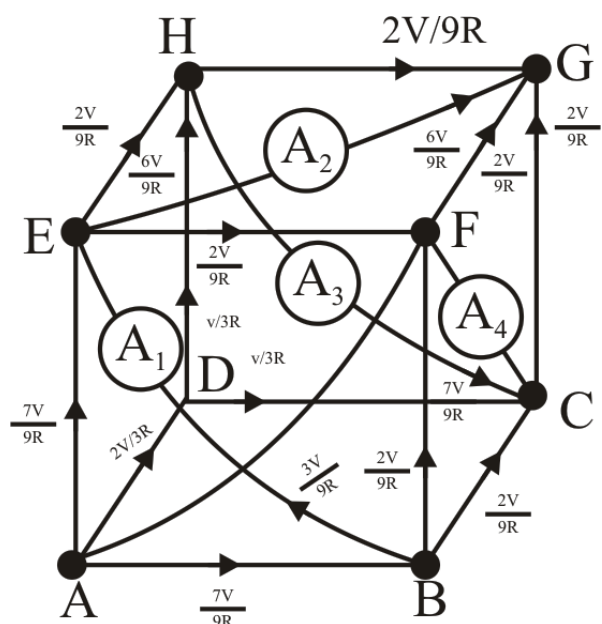


Current from A to D to (H, C) is $\overline{\left(R + \frac{R}{2}\right)}$

Current from A to E, B then to (H, C, F) is

$$\frac{V}{\frac{R}{2} + \frac{R}{7}} = \frac{14V}{9R}$$

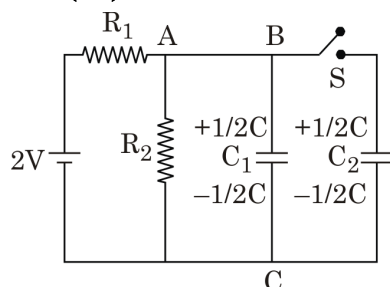
Now using KCL we can find current through each
 ammeter



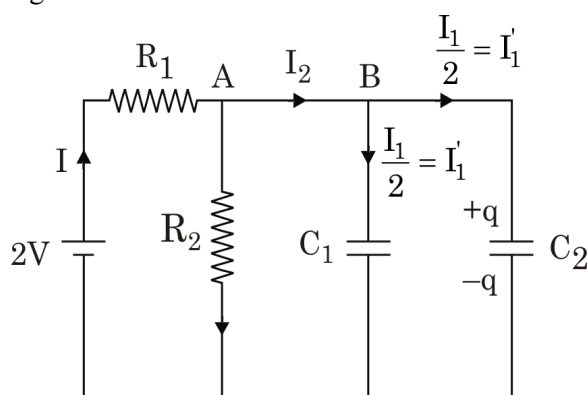
PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B)



When the switch is closed then at $t = 0$ the distribution of the charge is shown in the figure.



By applying Kirchoff's Law $I_1' = \frac{(1-q)}{2}$

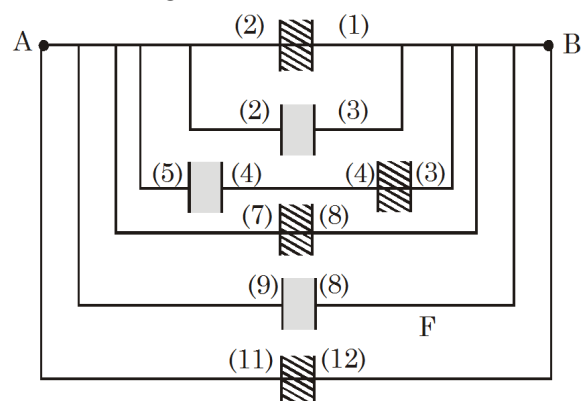
$$\int_{1/2}^q \frac{dq}{1-q} = \int_0^t \frac{dt}{2}$$

$$q = \left(1 - \frac{1}{2}e^{-t/2}\right)$$

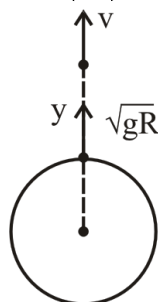
$$I_1' = \frac{dq}{dt} = \frac{1}{4}e^{-t/2}$$

8. Ans (D)

Apply Kirchoff's Law on the equivalent circuit shown in diagram.



9. Ans (A)



$$-\frac{GMm}{R} + \frac{1}{2}m(gR) = -\frac{GMm}{r} + \frac{1}{2}mv^2$$

$$-\frac{GMm}{R} + \frac{1}{2}m\frac{GMm}{R^2}R = -\frac{GMm}{r} + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = \frac{GMm}{r} - \frac{GMm}{2R}$$

$$\frac{v^2}{2} = \frac{GM}{r} - \frac{GM}{2R}$$

$$v^2 = \frac{2GM}{r} - \frac{GM}{R} \quad v = 0 \text{ at } r = 2R,$$

$$v = \sqrt{\frac{2GM}{r} - \frac{GM}{R}} = \frac{dr}{dt}$$

$$\int_{\pi/4}^0 \frac{2R(2\cos\theta)(-\sin\theta)d\theta}{\sqrt{\frac{1}{R}\tan^2\theta}} = \sqrt{GM} \cdot t$$

$$-\int_{\pi/4}^0 \frac{2R\sqrt{R} \cdot 2\cos\theta \sin\theta d\theta}{\left(\frac{\sin\theta}{\cos\theta}\right)} = \sqrt{GM}t$$

$$-\int_{\pi/4}^0 \cos^2\theta d\theta = \frac{\sqrt{GM}}{4R\sqrt{R}}t$$

$$\text{Let } \frac{2}{r} = \frac{1}{R}\sec^2\theta$$

$$r = 2R \cos^2\theta$$

$$dr = 2R(2\cos\theta)(-\sin\theta)d\theta$$

$$\text{at } r = R,$$

$$\frac{2}{R} = \frac{1}{R}\sec^2\theta$$

$$\sec\theta = \sqrt{2}$$

$$\theta = \frac{\pi}{4}$$

$$\text{at } r = 2R,$$

$$\frac{2}{2R} = \frac{1}{R}\sec^2\theta$$

$$\sec^2\theta = 1$$

$$\theta = 0$$

$$\text{at } r = \frac{3R}{2}$$

$$\frac{2}{\left(\frac{3R}{2}\right)} = \frac{1}{R} \sec^2 \theta$$

$$\frac{4}{3} = \sec^2 \theta$$

$$\sec \theta = \frac{2}{\sqrt{3}}$$

$$-\frac{1}{2} \int_{\frac{\pi}{4}}^0 (1 + \cos 2\theta) d\theta = \frac{\sqrt{GM}}{4R\sqrt{R}} t \quad \theta = \frac{\pi}{6}$$

$$-\frac{1}{2} \left[\left(0 - \frac{\pi}{4}\right) + \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{4}}^0 = \frac{\sqrt{GM}}{4R^{\frac{3}{2}}} \cdot t$$

$$-\frac{1}{2} \left[-\frac{\pi}{4} + \frac{1}{2}(0 - 1) \right] = \frac{\sqrt{GM}}{4R^{\frac{3}{2}}} t$$

$$\frac{1}{4} \left[\frac{\pi}{2} + 1 \right] = \frac{\sqrt{GM}}{4R^{\frac{3}{2}}} t$$

$$\left(\frac{\pi}{2} + 1\right) \frac{R^{\frac{3}{2}}}{\sqrt{GM}} = t$$

$$t = \left(\frac{\pi}{2} + 1\right) \sqrt{\frac{R^3}{GM}}$$

$$\int_R^{\frac{3R}{2}} \frac{dr}{\sqrt{\frac{2}{r} - \frac{1}{R}}} = \int_0^t \sqrt{GM} dt$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{6}} \frac{4R \cos \theta (-\sin \theta) d\theta}{\sqrt{\frac{2}{2R \cos^2 \theta} - \frac{1}{R}}} = \sqrt{GM} t$$

$$-\frac{4R}{\frac{1}{\sqrt{R}}} \int_{\frac{\pi}{4}}^{\frac{\pi}{6}} \frac{\cos \theta \sin \theta d\theta}{\tan \theta} = \sqrt{GM} t$$

$$-4R^{\frac{3}{2}} \int_{\frac{\pi}{4}}^{\frac{\pi}{6}} \cos^2 \theta d\theta = \sqrt{GM} t$$

$$-\frac{4R^{\frac{3}{2}}}{2} \left[\int_{\frac{\pi}{4}}^{\frac{\pi}{6}} (1 + \cos 2\theta) d\theta \right] = \sqrt{GM} t$$

$$-\frac{4R^{\frac{3}{2}}}{2} \left[\left(\frac{\pi}{6} - \frac{\pi}{4}\right) + \frac{1}{2} \sin 2\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{6}} = \sqrt{GM} t$$

$$\text{Let } \frac{2}{r} = \frac{1}{R} \sec^2 \theta$$

$$r = 2R \cos^2 \theta$$

$$dr = 2R (2 \cos \theta) (-\sin \theta) d\theta$$

$$dr = 4R \cos \theta (-\sin \theta) d\theta$$

$$\text{at } r = R, \cos^2 \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{4}$$

$$\text{at } r = \frac{3R}{2},$$

$$\frac{3R}{2} = 2R \cos^2 \theta$$

$$\cos^2 \theta = \frac{3}{4}$$

$$\theta = \frac{\pi}{6}$$

$$-\frac{4R^{\frac{3}{2}}}{2} \left[\frac{-2\pi}{24} + \frac{1}{2} \left(\frac{\sqrt{3}}{2} - 1 \right) \right] = \sqrt{GM} t$$

$$-\frac{4R^{\frac{3}{2}}}{2} \left[\frac{-\pi}{12} + \frac{1}{2} \left(\frac{\sqrt{3}}{2} - 1 \right) \right] = \sqrt{GM} t$$

$$\frac{4R^{\frac{3}{2}}}{2} \left[\frac{\pi}{12} - \frac{1}{2} \left(\frac{\sqrt{3}}{2} - 1 \right) \right] = \sqrt{GM} t$$

$$R^{\frac{3}{2}} \left[\frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1 \right] = \sqrt{GM} t$$

$$t = \frac{R^{\frac{3}{2}}}{\sqrt{GM}} \left[\frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1 \right]$$

$$-4R^{\frac{3}{2}} \int_{\frac{\pi}{6}}^0 \cos^2 \theta d\theta = \sqrt{GM} t$$

$$-\frac{4R^{\frac{3}{2}}}{2} \left[\int_{\frac{\pi}{6}}^0 d\theta + \int_{\frac{\pi}{6}}^0 \cos 2\theta d\theta \right] = \sqrt{GM} t$$

$$-2R^{\frac{3}{2}} \left[\left(0 - \frac{\pi}{6}\right) + \frac{\sin 2\theta}{2} \right]_{\frac{\pi}{6}}^0 = \sqrt{GM} t$$

$$-2R^{\frac{3}{2}} \left[-\frac{\pi}{6} + \frac{1}{2} \left(0 - \frac{\sqrt{3}}{2}\right) \right] = \sqrt{GM} t$$

$$-2R^{\frac{3}{2}} \left[-\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right] = \sqrt{GM} t$$

$$R^{\frac{3}{2}} \left[\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right] = \sqrt{GM} t$$

$$t = \frac{R^{\frac{3}{2}}}{\sqrt{GM}} \left[\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right]$$

$$T' = \frac{R^{\frac{3}{2}}}{\sqrt{GM}} \left[\frac{\pi}{3} + \frac{\sqrt{3}}{2} + \frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1 \right]$$

$$= \frac{R^{\frac{3}{2}}}{\sqrt{GM}} \left(\frac{\pi}{2} + 1 \right) t = \frac{R^{\frac{3}{2}}}{\sqrt{GM}} \left[\frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1 \right]$$

(A) $\alpha = 1$

(B) $\beta = \left(\frac{\pi}{3} - \sqrt{3} + 2 \right)$

(C) $\gamma = \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right)$

(D) $\delta = 1$

10. **Ans (D)**

$$V = -x^2y + 5$$

$$V(5, 2, 1) = -45V$$

$$V(1, 1, 1) = +4V$$

$$V(2, 1, 1) = 1V$$

$$V(1, 2, 1) = 3V$$

PART-1 : PHYSICS

SECTION-II

3. **Ans (37.50)**

$$\text{Force on dielectric slab} = F = \frac{1}{2} \frac{\epsilon_0 b}{d} V^2 (k-1)$$

block is moving constant velocity

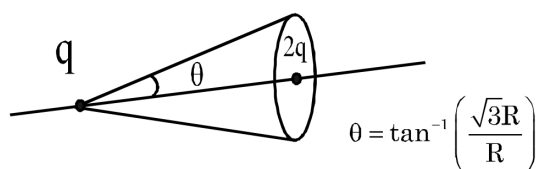
$$mg = F$$

$$mg = \frac{1}{2} \frac{\epsilon_0 b}{d} V^2 (k-1)$$

$$mg = \frac{1}{2} \times 10 \times (5)^2 (4-1)$$

$$m = \frac{25}{2} \times 3 = 37.5 \text{ kg}$$

4. **Ans (1.25)**

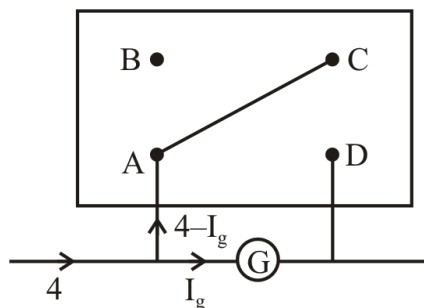


$$\phi_{\text{Total}} = \phi_{\text{due to } q} + \phi_{\text{due to } 2q}$$

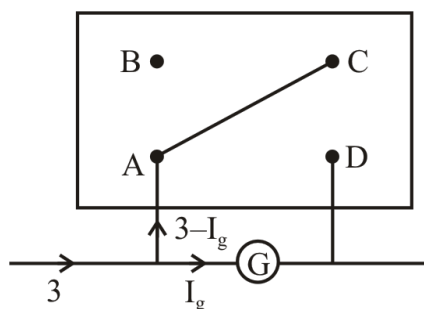
$$= \frac{q}{\epsilon_0} \frac{\left(1 - \frac{1}{2}\right)}{2} + \frac{1}{2} \left(\frac{2q}{\epsilon_0} \right)$$

$$= \frac{5q}{4\epsilon_0}$$

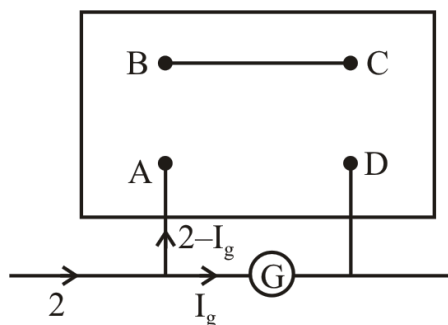
6. **Ans (0.58 to 0.59)**



$$I_g (R_g) = (4 - I_g) R_{CD}$$



$$I_g R_g = (3 - I_g) R_{AB}$$



$$I_g R_g = (2 - I_g) (R_{AB} + R_{CD})$$

$$I_g R_g = (2 - I_g) \left(\frac{I_g R_g}{3 - I_g} + \frac{I_g R_g}{4 - I_g} \right)$$

$$1 = (2 - I_g) \frac{(7 - 2I_g)}{(3 - I_g)(4 - I_g)}$$

$$12 + I_g^2 - 7I_g = 14 + 2I_g^2 - 11I_g$$

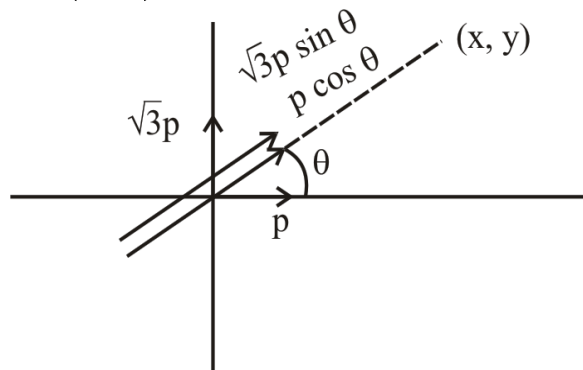
$$I_g^2 - 4I_g + 2 = 0$$

$$I_g^2 - 4I_g + 4 - 2 = 0$$

$$(I_g - 2)^2 = 0$$

$$I_g = 2 \pm \sqrt{2} = 0.58A$$

7. Ans (1.73)



$$v = \frac{kp \cos \theta}{R^2} + \frac{\sqrt{3}kp \sin \theta}{R^2}$$

$$= \frac{kp}{R^2} (\cos \theta + \sqrt{3} \sin \theta) = \frac{2kp}{R^2} \sin \left(\theta + \frac{\pi}{6} \right)$$

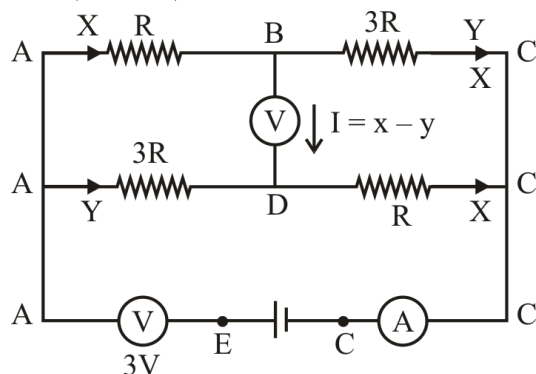
for V to be maximum, $\theta + \frac{\pi}{6} = \frac{\pi}{2}$

$$\theta = \frac{\pi}{3}$$

$$x = R \cos \frac{\pi}{3} = \frac{R}{2}$$

$$y = R \sin \frac{\pi}{3} = \frac{\sqrt{3}R}{2}$$

8. Ans (125.00)



$$x + y = 3.5 \Rightarrow x = 3.5 \text{ mA}$$

$$x = 1 + y \quad y = 2.5 \text{ mA}$$

KVL :

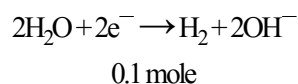
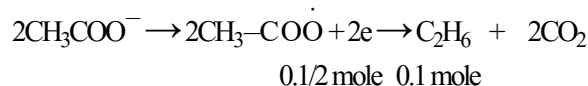
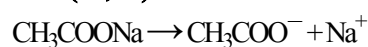
$$-3.5R - 500 + 2.5 \times 3R = 0$$

$$\Rightarrow 4R = 500 \Rightarrow R = 125\Omega$$

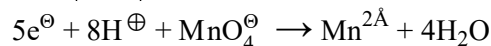
PART-2 : CHEMISTRY

SECTION-I (i)

2. Ans (B,D)



3. Ans (B,C)



$$E_{\text{MnO}_4^-/\text{Mn}^{2+}} = E_{\text{MnO}_4^-/\text{Mn}^{2+}}^0 - \frac{0.059}{5} \log \frac{1}{[\text{H}^+]^8}$$

$$E_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51 - \frac{0.059}{5} \log \frac{1}{(10^{-3})^8}$$

$$E_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.22 \text{ volt}$$

Now MnO_4^- can oxidise ion $\text{X}^\ominus$ then

$$\Rightarrow E_{\text{MnO}_4^-/\text{Mn}^{2+}} + E_{\text{X}^\ominus/\text{X}_2} > 0$$

$$\Rightarrow E_{\text{MnO}_4^-/\text{Mn}^{2+}} > E_{\text{X}_2/\text{X}^\ominus}$$

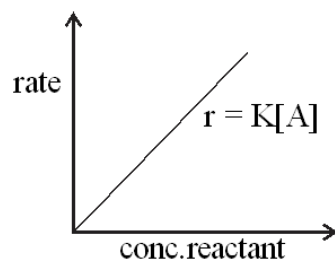
This condition is satisfied by $\text{I}^\ominus$ & $\text{Br}^\ominus$.

4. Ans (A,B,C)

(A) Rate law may contain reactants, products and sometimes catalysts.

(B) 1st order reaction may be elementary or complex

(C) for 1st order reaction; $\text{A} \rightarrow \text{P}$



for 1st order reaction, $t_{1/2} = \frac{\ln 2}{K}$ (independent of initial conc.)

5. Ans (A,C,D)

Difference in refractive indices of DM and DP cause easy detection by ultramicroscope.

Less number of particles gives less (negligible) colligative properties.

6. Ans (B,D)

$$\text{Average rate of reaction} = -\frac{1}{3} \left(\frac{\Delta P_{\text{H}_2}}{\Delta t} \right)$$

$$= \frac{1}{3} \left(\frac{380/760}{100} \right) = \frac{5}{3000} \text{ atm/sec}$$

$$\text{Average rate of formation of } \text{NH}_3(\text{g}) = \frac{1}{300} \text{ atm/sec}$$

$$\frac{\text{Rate}_{\text{N}_2}}{1} = \frac{\text{Rate}_{\text{NH}_3}}{2}$$

$$\frac{\text{Rate}_{\text{N}_2}}{1} = \frac{\text{Rate}_{\text{H}_2}}{3}$$

$$\log \left(\frac{-\Delta P_{\text{H}_2}}{\Delta t} \right) = \log \left(\frac{-\Delta P_{\text{N}_2}}{\Delta t} \right) + \log 3$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (D)**

Reaction



$$E = 0.00\text{V} - \frac{0.06}{2} \log \frac{[\text{H}^+]^2}{\text{pH}_2}$$

$$\text{pH}_2 = 1 \text{ atm}$$

$$E = -0.059 \log [\text{H}^+]$$

$$= 0.059 \times \text{pH}$$

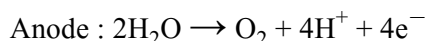
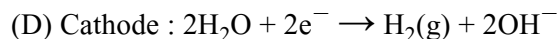
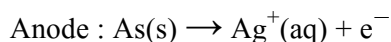
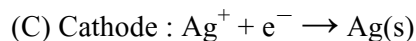
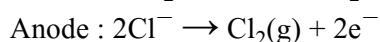
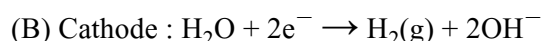
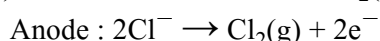
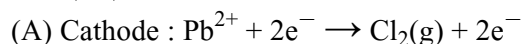
$$(a) \text{pH} = 7 \text{ (salt does not hydrolyse)} ; E = 0.42 \text{ V}$$

$$(b) \text{pH} = 7 - \frac{1}{2} [\text{pK}_b + \log c] = 4 ; E = 0.24 \text{ V}$$

$$(c) \text{pH} = 7 + \frac{1}{2} [\text{pK}_a + \log c] = 10 ; E = 0.6 \text{ V}$$

$$(d) \text{pH} = \frac{1}{2} [\text{pK}_{a1} + \text{pK}_{a2}] = 7 ; E = 0.42 \text{ V}$$

8. **Ans (A)**



9. **Ans (A)**

$$(P) \text{rate} = \frac{k[\text{x}]}{x_s + [\text{x}]} = \frac{k}{\frac{x_s}{[\text{x}]} + 1}$$

$$\text{If } [\text{x}] \rightarrow \infty \Rightarrow \text{rate} \rightarrow k \Rightarrow \text{order} = 0$$

$$\Rightarrow (P) - (3)$$

$$(Q) [\text{x}] \ll x_s \Rightarrow \text{rate} = \frac{k[\text{x}]}{x_s} \Rightarrow \text{order} = 1$$

$$\Rightarrow (Q) - (2), (5)$$

$$(R) [\text{x}] \gg x_s \Rightarrow \text{rate} = k \Rightarrow \text{order} = 0$$

$$\Rightarrow (R) - (1), (4)$$

$$(S) \text{rate} = \frac{k[\text{x}]^2}{x_s + [\text{x}]}$$

$$[\text{x}] \gg x_s \Rightarrow \text{rate} = k[\text{x}]$$

$$\Rightarrow (S) - (2), (5)$$

10. **Ans (D)**

(A) Precipitation of colloidal sol is known as coagulation.

(B) Purification of colloids is done by dialysis, electrodialysis and ultrafiltration

(C) Protecting power $\propto \frac{1}{\text{Gold number}}$

(D) Peptization is formation of colloidal sol from a precipitate by shaking it with dispersion medium in presence of small amount of electrolyte.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (0.25)**

1. Osmotic pressure should be lesser.

2. It's negative colloid

3. $\Delta H > 0$, $\Delta S > 0$, miscellisation occurs due to positive change in entropy

4. Statement corresponds to electro osmosis

5. Gold sol is hydrophobic sol

6. True statement

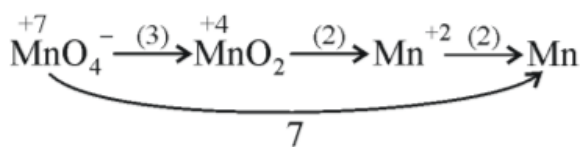
7. Starch sol stabilises gold sol.

8. Decrease in residual forces

9. Surface area changes, adsorption changes.

10. Conductivity does not change much initially, after all KI is consumed, conductivity increases

2. **Ans (0.49)**



For the required reaction $\Delta G^\circ = \Delta G^\circ_1 + \Delta G^\circ_2 + \Delta G^\circ_3$

$$\Rightarrow 7 \times E = 1.7 \times 3 + 1.25 \times 2 + (-1) \times 2$$

$$E = \frac{5.6}{7} = 0.8$$

Similarly for "Y"

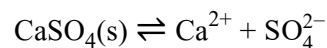
$$E = 1.86/6 = 0.31$$

$$\text{Ans.} = 0.49$$

3. Ans (6.00)

$$\Lambda_{\text{Na}_2\text{SO}_4} = \frac{2.6 \times 10^{-4}}{10^{-3}} \times 1000 = 260$$

$$\Lambda_{\text{SO}_4^{2-}} = 260 - 2(50) = 160$$



$$x \quad 10^{-3} + x$$

$$8.2 \times 10^{-4} = 10^{-3} [2 \times 10^{-3} \times 50 + x \times 120 +$$

$$(10^{-3} + x) \times 160]$$

$$\therefore x = 2 \times 10^{-3}$$

$$\text{Hence, } K_{\text{sp}} \text{ of } \text{CaSO}_4 = x \times (10^{-3} + x)$$

$$= 6 \times 10^{-6} \text{ M}^2$$

4. Ans (120.00)

For rate to reach 36%

$$A_t = 0.6 \text{ Å}$$

$$t = \frac{1}{k} \left[\frac{1}{0.6\text{Å}} - \frac{1}{\text{Å}} \right]$$

$$t = \frac{1}{k} \left[\frac{2}{3\text{Å}} \right]$$

$$k = \frac{1}{1440 \times 7}$$

$$\therefore t = 1440 \times 7 \left[\frac{2}{3} \times \frac{1}{56 \times 10^{-3}} \right]$$

$$t = 120 \times 10^3 \text{ sec}$$

5. Ans (15.00)

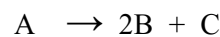
$$i \times t = (10 \times 100 + 10 \times 100) \times 10^{-3} \text{ A-sec.}$$

$$i \times t = 2A$$

$$30 = z \times 2$$

$$z = 15$$

7. Ans (52.50)



$$t = 0 \quad C_0$$

$$t = 20 \quad (C_0 - x_1) \quad 2x_1 \quad x_1$$

$$t = 50 \quad (C_0 - x_2) \quad 2x_2 \quad x_2$$

$\therefore$ only A is optically active

$$R_{20\text{min}} = \alpha_A \cdot (C_0 - x_1) = 30^\circ$$

$$R_{50\text{min}} = \alpha_A (C_0 - x_2) = 15^\circ$$

$$R_\infty = 0$$

$$k = \frac{1}{t_2 - t_1} \ln \frac{R_\infty - R_{20}}{R_\infty - R_{50}} = \frac{1}{30} \ln 2$$

$$t_{1/2} = \frac{\ln 2}{k} = 30\text{min}$$

now : let initial concentration of A = $[A]_0$ after

$$90 \text{ min } [k] = \frac{[A]_0}{2^3} = \frac{[A]_0}{8}$$

$$\Rightarrow [B] = 2 \left([A]_0 - \frac{[A]_0}{8} \right) = \frac{7[A]_0}{8} \times 2$$

$$[B]_{30\text{min}} = \left(\frac{[A]_0}{2} \right) \times 2 = [A]_0$$

$$[B]_{90\text{min}} = \frac{2 \times 7}{8} \times [A]_0$$

$$\frac{V_{\text{H}_2\text{O}_2, 30\text{min}}}{V_{\text{H}_2\text{O}_2, 90\text{min}}} = \frac{[A]_0}{\frac{14}{8}[A]_0}$$

$$\Rightarrow V_{\text{H}_2\text{O}_2, 90\text{min}} = \frac{14}{8} \times V_{\text{H}_2\text{O}_2(30\text{min})} = \frac{14}{8} \times 30 = 52.5\text{mL}$$

8. Ans (3.00)

$$\text{Number of mole of } H_2 = \frac{0.03 \times 2.46 \times 10^{-3}}{0.0820 \times 300}$$

$$= 3 \times 10^{-6} \text{ moles}$$

$$= 18 \times 10^{17} H_2 \text{ particles}$$

No. of sites available

$$= \frac{10^3 \times 10}{100} \times 6 \times 10^{15} = 6 \times 10^{17}$$

No. of molecules adsorbed per site :

$$= \frac{18 \times 10^{17}}{6 \times 10^{17}} = 3$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C,D)

$$\text{Let } f(x) = 1 \quad 0 \leq x < a$$

$$-1 \quad a < x < b$$

$$1 \quad b < x \leq 1$$

$$\text{As } \int_0^1 f(x) dx = 0 \Rightarrow a - (b - a) + (1 - b) = 0 ;$$

$$2a + 1 = 2b$$

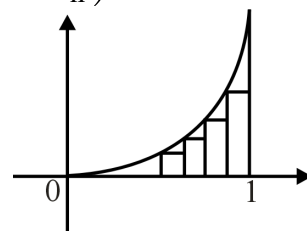
$$\begin{aligned} \int_0^1 xf(x) dx \\ &= \int_0^a xf(x) dx + \int_a^b xf(x) dx + \int_b^1 xf(x) dx \\ &= \frac{a^2}{2} - \left(\frac{b^2 - a^2}{2} \right) + \frac{1 - b^2}{2} = \frac{2a^2 - 2b^2 + 1}{2} \end{aligned}$$

$$I = \frac{2a^2 - 2\left(\frac{2a+1}{2}\right)^2 + 1}{2} = \frac{1 - 4a}{4} ; I_{\max} = \frac{1}{4}$$

$$\text{similarly minimum is } -\frac{1}{4}$$

2. Ans (A,C,D)

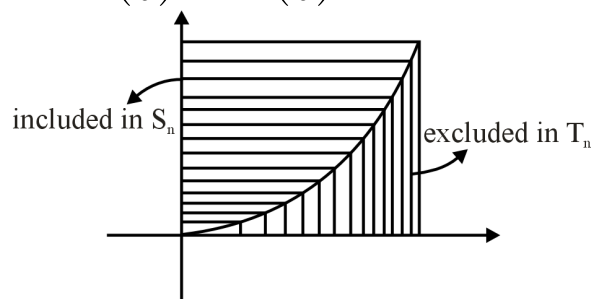
$$S_n = \left(\frac{1}{n^6} + \frac{32}{n^6} + \frac{243}{n^6} + \dots + \frac{1}{n} \right)$$



$$T_n = \left(\frac{1}{n^6} + \frac{32}{n^6} + \frac{243}{n^6} + \dots + \frac{(n-1)^5}{n^6} \right)$$

$$S_\infty = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r^5}{n^6} = \int_0^1 x^5 dx = \frac{x^6}{6} \Big|_0^1 = \frac{1}{6}$$

$$\Rightarrow S_n > \left(\frac{1}{6} \right) ; T_n < \left(\frac{1}{6} \right)$$



But since f is concave upward the area included in S_n , for any two consecutive values of r is more than area excluded in T_n , for same values of r .

3. Ans (A,C)

$$\text{Put } y^2 = u$$

$$\begin{aligned} 2y \frac{dy}{dx} &= \frac{du}{dx} \\ \frac{du}{dx} - \frac{u}{x} &= \frac{1}{x} + x \end{aligned}$$

Use LDE

$$\text{I.F.} = e^{-\int \frac{1}{x} dx} = \frac{1}{x}$$

$$u \cdot \frac{1}{x} = \int \left(\frac{1}{x} + x \right) \left(\frac{1}{x} \right) dx$$

$$\frac{y^2}{x} = \frac{-1}{x} + x + c$$

$$y(1) = 1, c = 1$$

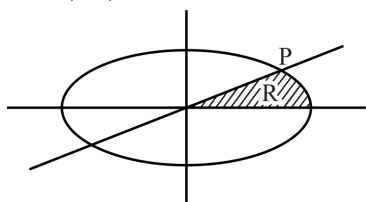
$$y^2 = -1 + x^2 + x$$

which is hyperbola

4. Ans (A,D)

$$\begin{aligned} \int_2^3 \frac{e^{x-1} \cdot 2x}{(x-1)(x-4)} dx &= \int_2^3 e^{x-1} \left[\frac{-2/3}{x-1} + \frac{8/3}{x-4} \right] dx \\ &= -\frac{2}{3} \int_2^3 \frac{e^{x-1}}{x-1} dx + \frac{8}{3} \int_2^3 \frac{e^{x-1}}{x-4} dx \\ &= -\frac{2}{3} \int_1^2 \frac{e^x}{x} dx + \frac{8e^3}{3} \int_{-2}^{-1} \frac{e^x}{x} dx \end{aligned}$$

5. Ans (A)



Stretch the figure by a factor 3 along the y-axis.
So that the point (x, y) goes to (x, 3y) then new line is $2y = 3x$

So, for $R = R'$, $m = \frac{2}{9}$

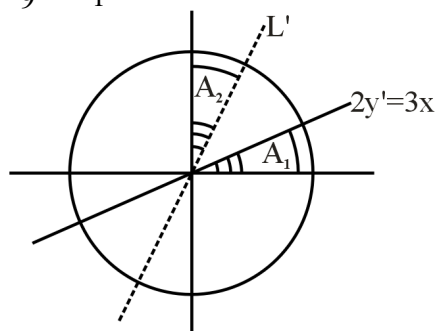
$$\frac{x^2}{9} + \frac{y^2}{1} = 1 ; 2y = x$$

$$P \equiv \left(\frac{3}{\sqrt{3}}, \frac{6}{\sqrt{3}} \right)$$

Now replace y by $\frac{y'}{3}$, we will reach

$$2y = x \rightarrow 2y' = 3x$$

$$\frac{x^2}{9} + \frac{y^2}{1} = 1 \rightarrow x^2 + (y')^2 = 9$$



Let A_1 = Area bounded by new line & circle

Now $A_1 = 3R$ (use integration and see)

for $A_1 = A_2$

line $L' \equiv 3y' = 2x$

Now we will go to our old system by replacing

$y' \rightarrow 3y$

Hence $L' \equiv 3y' = 2x \rightarrow 9y = 2x$

$$m = \frac{2}{9}$$

6. Ans (A,C,D)

Differentiating the given equation $2f(x) f'(x)$

$$= (f(x))^2 + (f'(x))^2 \Rightarrow (f(x) - f'(x))^2 = 0$$

$$\Rightarrow f'(x) = f(x) \Rightarrow f(x) = ce^x$$

$$\therefore f(0) = \sqrt{K} \Rightarrow c = \sqrt{K} \Rightarrow f(x) = \sqrt{K} e^x$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

(I) put $t + 2 = z$, $t = z - 2$

$$t \rightarrow z \quad dt = dz$$

$$z = x + 2$$

$$t \rightarrow 1, z = 3$$

$$\int_3^{x+2} \frac{e^{z-2}}{z} dz$$

$$e^{-2} \int_3^{x+2} \frac{e^z}{z} dz = e^{-2} \int_3^{x+2} \frac{e^t}{t} dt$$

$$F(x+2) - F(3) = \int_1^{x+2} \frac{e^t}{t} dt - \int_1^3 \frac{e^t}{t} dt$$

$$= \int_1^{x+2} \frac{e^t}{t} dt + \int_3^1 \frac{e^t}{t} dt$$

$$= \int_3^{x+2} \frac{e^t}{t} dt$$

$$\Rightarrow e^{-2} \int_3^{x+2} \frac{e^t}{t} dt = e^{-2} (F(x+2) - F(3))$$

Similarly in

(II) put $3t = z$

(III) Integrate by part

(IV) put $\frac{1}{t} = z$

8. Ans (B)

$$(I) \int \frac{x^4 - 1}{x^3 \sqrt{x^4 + x^2 + 1/x}} dx,$$

$$\text{Let } \frac{\sqrt{x^4 + x^2 + 1}}{x} = t$$

$$\int \frac{x^2 \cdot \sqrt{x^4 + x^2 + 1}}{x^3 \cdot t} dt,$$

$$\left(\frac{x^4 - 1}{x^2 \cdot \sqrt{x^4 + x^2 + 1}} \right) dx = dt$$

$$\int \frac{\sqrt{x^4 + x^2 + 1}}{x \cdot t} dt,$$

$$(x^4 - 1)dx = x^2 \cdot \sqrt{x^4 + x^2 + 1} dt$$

$$\int \frac{t}{t} dt, \int dt = t + C = \frac{\sqrt{x^4 + x^2 + 1}}{x} + C$$

$$(II) \int \frac{(x^2 - 1)}{x\sqrt{1+x^4}} dx, \int \frac{(x^2 - 1)}{x^2\sqrt{x^2 + \frac{1}{x^2}}} dx,$$

$$\text{Let } x + \frac{1}{x} = t, \left(1 - \frac{1}{x^2}\right) dx = dt$$

$$\int \frac{dt}{\sqrt{t^2 - 2}} = \ln(t + \sqrt{t^2 - 2}) + C$$

$$= \ln\left(x + \frac{1}{x} + \sqrt{\left(x + \frac{1}{x}\right)^2 - 2}\right) + C$$

$$\ln\left(\frac{(x^2 + 1) + \sqrt{1 + x^4}}{x}\right) + C$$

$$(III) \int \frac{(1+x^2)}{(1+x^2)\sqrt{1+x^4}} dx = \int \frac{\left(1 + \frac{1}{x^2}\right) dx}{x\left(1 - \frac{1}{x^2}\right)\sqrt{\frac{1}{x^2} + x^2}}$$

$$= \int \frac{\left(1 + \frac{1}{x^2}\right) dx}{\left(x - \frac{1}{x}\right) \sqrt{\left(x - \frac{1}{x}\right)^2 + 2}}$$

$$\text{Let } \left(x - \frac{1}{x}\right) = t, \left(1 + \frac{1}{x^2}\right) dx = dt, \int \frac{dt}{t\sqrt{t^2 + 2}}$$

$$\text{Let } t^2 + 2 = y^2, 2t dt = 2y dy$$

$$\int \frac{dy}{t^2} = \int \frac{dy}{(y^2 - 2)} = \frac{1}{2\sqrt{2}} \ln\left(\frac{y - \sqrt{2}}{y + \sqrt{2}}\right) + C$$

$$= \frac{1}{2\sqrt{2}} \ln\left(\frac{\sqrt{t^2 + 2} - \sqrt{2}}{\sqrt{t^2 + 2} + \sqrt{2}}\right) + C$$

$$= \frac{1}{2\sqrt{2}} \ln\left(\frac{(\sqrt{t^2 + 2} - \sqrt{2})^2}{t^2}\right) + C$$

$$= \frac{1}{\sqrt{2}} \ln\left(\frac{\sqrt{t^2 + 2} - \sqrt{2}}{t}\right) + C$$

$$= \frac{1}{\sqrt{2}} \ln\left(\frac{\sqrt{\left(x - \frac{1}{x}\right)^2 + 2} - \sqrt{2}}{\left(x - \frac{1}{x}\right)}\right) + C$$

$$= \frac{1}{\sqrt{2}} \ln\left(\frac{\sqrt{x^4 + 1} - \sqrt{2}x}{(x^2 - 1)}\right) + C$$

$$(IV) \int \frac{dx}{(1+x^4)\sqrt{\sqrt{1+x^4} - x^2}}$$

$$\Rightarrow \int \frac{dx}{x^5 \left(1 + \frac{1}{x^4}\right) \sqrt{\sqrt{1 + \frac{1}{x^4}} - 1}}$$

$$\text{Let } \sqrt{1 + \frac{1}{x^4}} - 1 = t^2$$

$$\Rightarrow \frac{1}{2\sqrt{1 + \frac{1}{x^4}}} \left(\frac{-4}{x^5}\right) dx = 2t dt$$

$$\Rightarrow \frac{dx}{x^5} = -t \sqrt{1 + \frac{1}{x^4}} dt$$

$$\Rightarrow \int \frac{-t \sqrt{1 + \frac{1}{x^4}} dt}{\left(1 + \frac{1}{x^4}\right) t} \Rightarrow - \int \frac{dt}{\sqrt{1 + \frac{1}{x^4}}}$$

$$\Rightarrow - \int \frac{dt}{(t^2 + 1)} \Rightarrow -\tan^{-1} t + C$$

$$\Rightarrow C - \tan^{-1} \left(\sqrt{\sqrt{1 + \frac{1}{x^4}} - 1}\right)$$

$$C - \tan^{-1} \left(\sqrt{\sqrt{1 + \frac{1}{x^4}} - 1}\right)$$

9. Ans (C)

$$(I) \quad xdy = ydx + y^2 dy \Rightarrow \frac{xdy - ydx}{y^2} = dy$$

$$\Rightarrow -d\left(\frac{x}{y}\right) = dy$$

$$-\frac{x}{y} = y + c \text{ put } x = 1, y = 1 \Rightarrow c = -2$$

$$-\frac{x}{y} = y - 2 \Rightarrow \frac{x}{3} = -5$$

$$(II) \quad \frac{dy}{dt} - \frac{t}{t+1}y = \frac{1}{t+1}$$

$$\text{I.f. } e^{-\int \frac{t+1-1}{t+1} dt} = e^{-t+\ln(t+1)} = (t+1)e^{-t}$$

$$\text{Solution is } (t+1)e^{-t}y = -e^{-t} + C \text{ put}$$

$$t = 0 \text{ and } y = -1 \Rightarrow c = 0$$

$$\therefore 2e^{-1}y = -e^{-1}$$

$$y = -\frac{1}{2}$$

$$(III) \quad (x^2 + y^2)dy = xydx$$

$$\frac{dy}{dx} = \frac{xy}{x^2 + y^2} \text{ put } y = vx, \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow \ln v - \frac{1}{2v^2} = -\ln x + c$$

$$\therefore c = -\frac{1}{2} \therefore \ln \frac{y}{x} - \frac{1}{2} \frac{x^2}{y^2} = -\ln x - \frac{1}{2}$$

$$\text{put } y = e$$

$$\therefore x = \sqrt{3}e$$

$$(IV) \quad \frac{dy}{dx} + 2\frac{y}{x} = 0$$

$$x^2y = C \text{ put } x = 1, y = 1 \text{ and we get } C = 1$$

$$\text{put } x = 2 \Rightarrow y = \frac{1}{4}$$

10. Ans (C)

$$(I) \quad \text{Area} = 4 \int_0^1 \left(2^{-x} - \frac{1}{2}\right) dx$$

$$= 4 \left[\frac{-2^{-x}}{\ln 2} - \frac{x}{2} \right]_0^1$$

$$= 4 \left[-\frac{1}{2\ln 2} - \frac{1}{2} + \frac{1}{\ln 2} \right]$$

$$= 4 \left[\frac{1}{2\ln 2} - \frac{1}{2} \right]$$

$$= \frac{2}{\ln 2} - 2$$

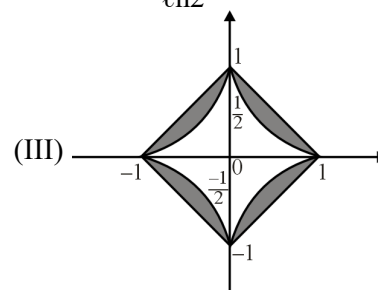
$$(II) \quad \text{Area} = 4 \left[\int_0^{\frac{1}{2}} \left(2^{-x} - \frac{1}{2}\right) dx \right]$$

$$= 4 \left[\frac{-2^{-x}}{\ln 2} - \frac{x}{2} \right]_0^{1/2}$$

$$= 4 \left[\frac{-1}{\sqrt{2}\ln 2} - \frac{1}{4} + \frac{1}{\ln 2} \right]$$

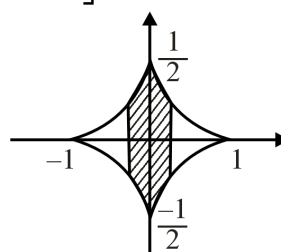
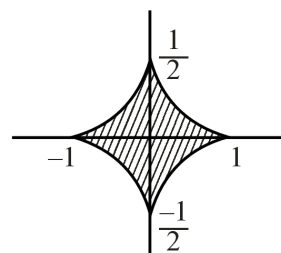
$$= \frac{4}{\ln 2} \left[1 - \frac{1}{\sqrt{2}} \right] - 1$$

$$= \frac{(\sqrt{2}-1)2\sqrt{2}}{\ln 2} - 1$$

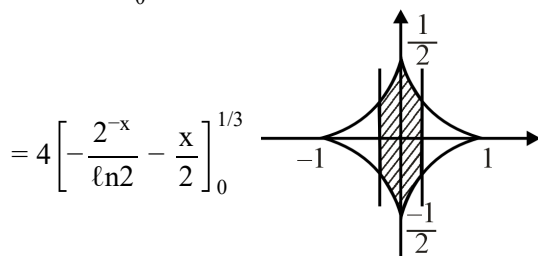


$$\text{Area} = 2 - \left(\frac{2}{\ln 2} - 2 \right)$$

$$= 4 - \frac{2}{\ln 2}$$



$$(IV) \text{ Area} = 4 \int_0^{1/3} \left(2^{-x} - \frac{1}{2} \right) dx$$



$$\begin{aligned} &= 4 \left[-\frac{1}{2^{1/3} \ln 2} - \frac{1}{6} + \frac{1}{\ln 2} \right] \\ &= \frac{4}{\ln 2} \left[1 - \frac{1}{2^{1/3}} \right] - \frac{2}{3} \\ &= \frac{4}{2^{1/3} \ln 2} (2^{1/3} - 1) - \frac{2}{3} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.25)

$$f(0) = 2 \Rightarrow f(x+2) + f(x-2) = f(x)$$

$$\Rightarrow f(x) = f(x+12)$$

$$\int_0^{12} f(x) dx = \frac{1}{4} \int_{-24}^{24} f(x-36) dx$$

2. Ans (3.00)

$$\sum_{r=0}^{\infty} \int_0^1 x^{4r+1} - x^{4r+2} dx$$

$$\int_0^1 \left(\frac{1}{1-x^4} \right) - \left(\frac{x^2}{1-x^4} \right) dx$$

$$\int_0^1 \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_0^1$$

$$\Rightarrow \frac{\pi}{4} = L$$

$$4L = \pi$$

$$[4L] = [\pi] = 3$$

3. Ans (16.00)

$$x \frac{d}{dx} (\ln(f(g(x)))) = \frac{d}{dx} (\ln(g(f(x))))$$

$$\int_0^p f(g(x)) dx = \frac{1}{2} - \frac{e^{-2p}}{2} \Rightarrow f(g(p)) = e^{-2p}$$

$$\therefore x \frac{d}{dx} \ln(e^{-2x}) = \frac{d}{dx} \ln(g(f(x)))$$

$$\Rightarrow -2x = \frac{d}{dx} (\ln(g(f(x))))$$

$$\Rightarrow \ln(g(f(x))) = -x^2 + c$$

$$g(f(x)) = k' \cdot e^{-x^2}; g(f(0)) = 1 \Rightarrow k' = 1;$$

$$g(f(4)) = e^{-16}$$

$$\Rightarrow k = 16$$

4. Ans (8.00)

$$y = 1 + 4x - x^2 = 5 - (x-2)^2$$

$$\Rightarrow (x-2)^2 = 5 - y = -(y-5). \text{ This is parabola}$$

with vertex A at (2, 5) and facing downwards.

$$\begin{aligned} \text{Area OMEAB} &= \int_0^{\frac{3}{2}} (1 + 4x - x^2) dx = \left[x + 2x^2 - \frac{x^3}{3} \right]_0^{\frac{3}{2}} \\ &= \frac{3}{2} + \frac{9}{2} - \frac{27}{24} = \frac{39}{8} \end{aligned}$$

OE is line $y = mx$, so that E is $\left(\frac{3}{2}, \frac{3m}{2} \right)$

$$\text{Area of } \Delta OME = \frac{1}{2} \times \frac{3}{2} \times \frac{3m}{2} = \frac{1}{2} \times \frac{39}{8} \text{ (By problem)}$$

$$\Rightarrow m = \frac{13}{6}$$

5. Ans (7.00)

$$I = \int \frac{x+2}{\sqrt{(x+2)^4 - 4(x+2)^2 + 4}} dx$$

$$\text{Let } (x+2)^2 = t \Rightarrow \frac{dt}{dx} = 2(x+2) \Rightarrow \frac{dt}{2} = (x+2)dx$$

$$\Rightarrow I = \frac{1}{2} \int \frac{1}{\sqrt{t^2 - 4t + 4}} dt$$

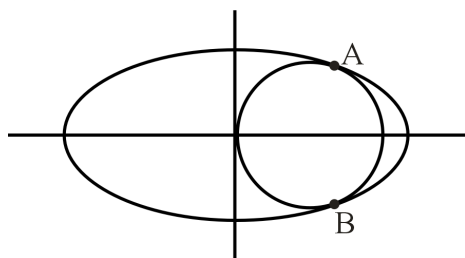
$$\Rightarrow I = \frac{1}{2} \int \frac{1}{t-2} dt = \frac{1}{2} \ln|t-2| + c$$

6. Ans (0.46 to 0.48)

Area of parallelogram formed by

$$\left\{ \begin{array}{l} 5x - y = 2 \\ 5x - y = -2 \\ x - 7y = 2 \\ x - 7y = -2 \end{array} \right\} \text{ equals } \frac{4}{17}$$

7. Ans (3.00)



E and C touch each other at A and B solve E

$$\text{and } C \equiv b^2x^2 + a^2[c^2 - (x - c)^2] = a^2b^2$$

$$\Rightarrow b^2x^2 + a^2[2cx - x^2] = a^2b^2$$

$$\Rightarrow (b^2 - a^2)x^2 + 2ca^2x - a^2b^2 = 0$$

$$D = 0 \Rightarrow 4c^2a^4 = 4a^2b^2(a^2 - b^2)$$

$$\Rightarrow a^2 = \frac{b^4}{b^2 - c^2}$$

$$\Delta = \pi ab = \frac{\pi b^2}{\sqrt{b^2 - c^2}} = \frac{\pi}{\sqrt{b^{-4} - c^2b^{-6}}} \frac{dA}{db} = 0$$

$$\Rightarrow 6b^{-7}c^2 = 4b^{-5} \Rightarrow b^2 = \frac{3c^2}{2}$$

$$b^2 = \frac{3c^2}{2} \text{ and } a^2 = \frac{9c^2}{2}$$

$$\therefore e = \frac{\sqrt{2}}{\sqrt{3}}, \Delta = \frac{3\sqrt{3}\pi}{2}c^2$$

8. Ans (4.00)

$$p(x) = \int_{-2}^x f(t)(x-t) dt + \int_x^2 f(t)(t-x) dx ;$$

$$p'(x) = \int_{-2}^x f(t) dt - \int_x^2 f(t) dt$$

$$p''(x) = f(x) + f(x) = 2f(x) ;$$

$$\int_0^2 \frac{p''(x)}{f(x)} dx = 2 \int_0^2 dx = 4$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

29-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	A	D	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,D	A,B	A,D	B,C	B,C	A,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	9	6	1	5	2	3	3	3

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	A	C	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B	A,C	A,C,D	A,D	B,D	A,B,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	7	3	5	4	9	6	3	5

PART-3 : MATHEMATICS

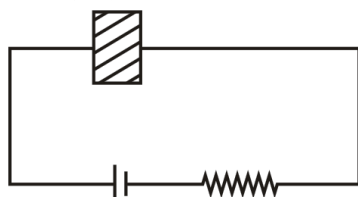
SECTION-I (i)	Q.	1	2	3	4				
	A.	C	D	C	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C	A,B,D	B,D	A,B,C	A,B,C	A,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	4	7	0	3	9	1	5	4

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

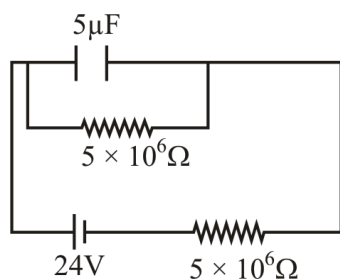


$$C = \frac{\epsilon_0 A}{d} = \frac{5 \times 8.85 \times 10^{-12} \times 1}{8.85 \times 10^{-6}}$$

$$C = 5 \times 10^{-6} = 5 \mu\text{F}$$

$$R = \frac{\rho \ell}{A} = \frac{5 \times 10^{12}}{8.85} \times \frac{8.85 \times 10^{-6}}{1}$$

$$= 5 \times 10^6 \Omega$$



$$i = \frac{24}{10 \times 10^6} = 2.4 \times 10^{-6} \text{ A}$$

$$\Delta V_C = 2.4 \times 10^{-6} \times 5 \times 10^6$$

$$= 12 \text{ V}$$

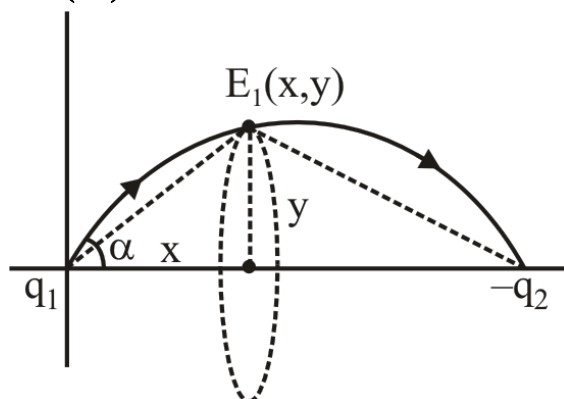
$$q_C = 5 \times 10^{-6} \times 12$$

$$= 60 \mu\text{C}$$

$$\tau = RC = 25 \text{ sec}$$

$$T_{1/2} = RC \ln 2 = (25 \ln 2) \text{ sec.}$$

2. Ans (A)



$$E_1 = \frac{kq_1}{x^2 + y^2}, E_2 = \frac{kq_2}{(\ell - x)^2 + y^2}$$

$$q_1 (1 - \cos \alpha) = \int E_1 \cos \alpha_1 (2\pi y dy) + \int E_2 \cos \alpha_2 (2\pi y dy)$$

$$= kq \left[\int \frac{1}{(x^2 + y^2)^{3/2}} \cdot x \cdot 2\pi y dy + \int \frac{1}{(\ell - x)^2} (\ell - x) (2\pi y dy) \right]$$

$$= kq \left[x (2\pi) \int_0^y \frac{y dy}{(x^2 + y^2)^{3/2}} + (2\pi) (\ell - x) \int_0^y \frac{y dy}{((\ell - x)^2 + y^2)^{3/2}} \right]$$

$$= k (2\pi) \left[x q_1 \left(\frac{1}{x} - \frac{1}{(x^2 + y^2)^{1/2}} \right) + (\ell - x) q_2 \left(\frac{1}{\ell - x} - \frac{1}{((\ell - x)^2 + y^2)^{1/2}} \right) \right]$$

$$= \frac{1}{2\epsilon_0} \left[q_1 \left(1 - \frac{x}{(x^2 + y^2)^{1/2}} \right) + q_2 \left(1 - \frac{(\ell - x)}{((\ell - x)^2 + y^2)^{1/2}} \right) \right]$$

$$q_1 (1 - \cos \alpha) = \frac{1}{2\epsilon_0} \left(q_1 \left(1 - \frac{x}{(x^2 + y^2)^{1/2}} \right) + q_2 \left(1 - \frac{\ell - x}{((\ell - x)^2 + y^2)^{1/2}} \right) \right)$$

3. Ans (D)

$$g' = g - \omega^2 R \cos^2 30^\circ = 0$$

4. Ans (A)

Both would produce $\frac{2k\lambda}{R}$ electric field and

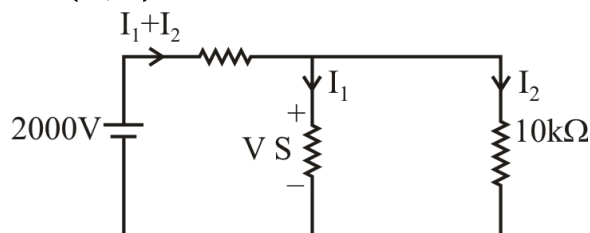
angle between field would be 60°

$$\therefore E_{\text{resultant}} = 2 \times \left(\frac{2k\lambda}{R} \right) \times \cos 30 = \frac{\sqrt{3}kQ}{\pi R^2}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,D)



$$I_2 (10 \times 10^3) = V$$

$$2000 - (I_1 + I_2) 5 \times 10^3 - V = 0$$

$$2000 - 5 \times 10^3 I_1 - \frac{V}{2} - V = 0$$

$$V = 10^3 \text{ V}$$

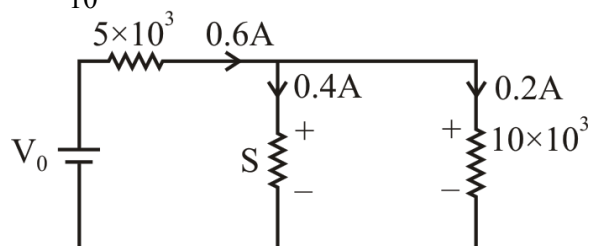
$$I_2 = 0.1 \text{ A}$$

$$2000 - 5 \times 10^3 \times 10^{-7} V^2 - \frac{3V}{2} = 0$$

$$5 \times 10^4 V^2 + \frac{3V}{2} - 2000 = 0$$

$$V = \frac{-\frac{3}{2} \pm \sqrt{\left(\frac{3}{2}\right)^2 + 4 \times 2000 \times 5 \times 10^{-4}}}{2 \times 5 \times 10^{-4}}$$

$$= \frac{-\frac{3}{2} \pm \sqrt{\frac{9}{4} + 4}}{10^{-3}} = \frac{-\frac{3}{2} + \frac{5}{2}}{10^{-3}} = 10^3 \text{ V}$$



$$V_0 - 5 \times 10^3 \times 0.6 - 0.2 \times 10 \times 10^3 = 0$$

$$V_0 = 3 \times 10^3 + 2 \times 10^3 = 5 \times 10^3 \text{ V}$$

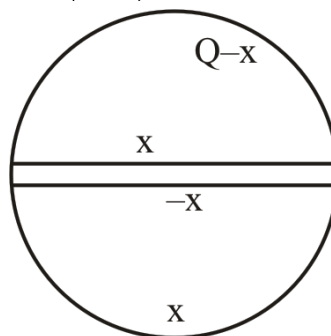
$$\Delta V = 0.2 \times 10 \times 10^3 = 2000 \text{ V}$$

$$I = 10^{-7} \times 2000^2$$

$$= 4 \times 10^{-7}$$

$$= 0.4 \text{ A}$$

6. Ans (A,B)

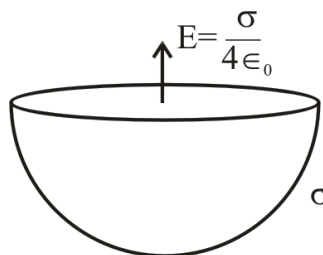


$$Q - x = x \Rightarrow x = \frac{Q}{2}$$

$$\sigma = \frac{\frac{Q}{2}}{2\pi R^2} = \frac{Q}{4\pi R^2}$$

$$\downarrow F = \frac{x^2}{2A\epsilon_0} = \frac{\left(\frac{Q}{2}\right)^2}{2\pi R^2 \epsilon_0} = \frac{Q^2}{8\pi \epsilon_0 R^2}$$

$$\downarrow F = \frac{\sigma}{4\epsilon_0} \cdot (-x) = \frac{Q}{4\pi R^2} \cdot \frac{1}{4\epsilon_0} \cdot \frac{Q}{2}$$



$$\downarrow F = \frac{Q^2}{32\pi \epsilon_0 R^2}$$

$$\uparrow F = \frac{\sigma}{4\epsilon_0} \cdot x = \frac{Q}{4\pi R^2} \cdot \frac{1}{4\epsilon_0} \cdot \frac{Q}{2}$$

$$\uparrow F = \frac{Q^2}{32\pi \epsilon_0 R^2}$$

$$\uparrow F = \frac{\sigma}{4\epsilon_0} \cdot (Q - x) = \frac{Q}{4\pi R^2} \cdot \frac{1}{4\epsilon_0} \cdot \frac{Q}{2}$$

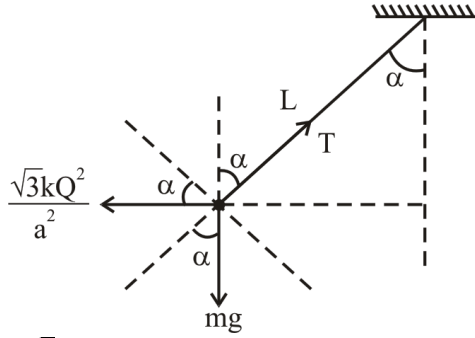
$$\uparrow F = \frac{Q^2}{32\pi \epsilon_0 R^2}$$

$$F_{\text{net}} = \frac{Q^2}{8\pi \epsilon_0 R^2} - \frac{Q^2}{32\pi \epsilon_0 R^2} = \frac{3Q^2}{32\pi \epsilon_0 R^2}$$

$$V_1 = \frac{kQ^2}{2R} = \frac{1}{4\pi \epsilon_0} \cdot \frac{Q^2}{2R} = \frac{Q^2}{8\pi \epsilon_0 R}$$

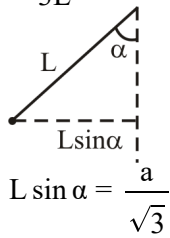
$$V_2 = \frac{Q^2}{2C} = \frac{\left(\frac{Q}{2}\right)^2 d}{2\epsilon_0 \pi R^2} = \frac{Q^2 d}{8\epsilon_0 \pi R^2}$$

7. Ans (A,D)



$$\frac{\sqrt{3}kQ^2}{a^2} \cos \alpha - mg \sin \alpha = ma_t$$

$$\frac{\sqrt{3}kQ^2}{3L^2} \times 4 \times \frac{\sqrt{3}}{2} - mg \frac{1}{2} = ma_t$$



$$L \sin \alpha = \frac{a}{\sqrt{3}}$$

$$\frac{L}{2} = \frac{a}{\sqrt{3}}$$

$$a = \frac{\sqrt{3}L}{2}$$

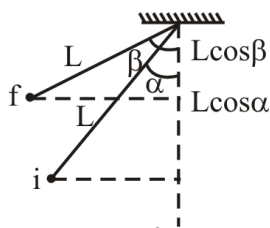
$$\frac{9 \times 10^9 \times 9 \times 10^{-14} \times 2 \times 10}{0.2 \times 0.2} - \frac{9 \times 10^{-4} \times 10}{2}$$

$$= 9 \times 10^{-4} a_t$$

$$\frac{81}{2} \times 10^{-3} - \frac{9}{2} \times 10^{-3} = 9 \times 10^{-4} a_t$$

$$36 \times 10^{-3} = 9 \times 10^{-4} a_t$$

$$a_t = 40 \text{ m/s}^2$$



$$v_i = \frac{3kQ^2}{\sqrt{3}L \sin \alpha} + 0$$

$$v_f = \frac{3kQ^2}{\sqrt{3}L \sin \beta} + 3mg(L \cos \alpha - L \cos \beta)$$

$$\frac{3kQ^2}{\sqrt{3}L \sin \alpha} = \frac{3kQ^2}{\sqrt{3}L \sin \beta} + 3mgL(\cos \alpha - \cos \beta)$$

$$\frac{3kQ^2}{\sqrt{3}L \times 1} = \frac{3kQ^2}{\sqrt{3}L \sin \beta} + 3mgL \left(\frac{\sqrt{3}}{2} - \cos \beta \right)$$

$$\frac{6}{\sqrt{3}} \times \frac{9 \times 10^9 \times 9 \times 10^{-14} \times 10}{0.2} - 9 \times 10^{-4} \times 10 \times \frac{0.2}{10} \times \frac{\sqrt{3}}{2} \times 3$$

$$= \frac{3kQ^2}{\sqrt{3}L \sin \beta} - 3mgL \cos \beta$$

$$81\sqrt{3} \times 10^{-4} - 27\sqrt{3} \times 10^{-4} = \frac{3 \times 9 \times 10^9 \times 9 \times 10^{-14} \times 10}{\sqrt{3} \times 0.2 \sin \beta}$$

$$3 \times 9 \times 10^{-4} \times 10 \times \frac{0.2}{10} \cos \beta$$

$$54\sqrt{3} \times 10^{-4} = \frac{81\sqrt{3}}{2} \times \frac{10^{-4}}{\sin \beta} - 6 \times 10^{-4} \cos \beta + \frac{1}{2} 3mv^2$$

$$6\sqrt{3} - \frac{9\sqrt{3}}{2 \sin \beta} + 6 \cos \beta = \frac{1}{2} \times 3 \times 9 \times 10^{-4} v^2$$

$$6\sqrt{3} - \frac{9\sqrt{3}}{2 \sin \beta} + 6 \cos \beta = \frac{3}{2} v^2$$

$$6\sqrt{3} - \frac{9\sqrt{3}}{2} \operatorname{cosec} \beta + 6 \cos \beta = \frac{3}{2} v^2$$

$$\text{for } v \text{ to be maximum, } \frac{d(v^2)}{d\beta} = 0$$

$$0 + \frac{9\sqrt{3}}{2} \cdot \operatorname{cosec} \beta \cdot \cot \beta + 6(-\sin \beta) = 0$$

$$\frac{9\sqrt{3}}{2} \cdot \frac{\cos \beta}{\sin^2 \beta} = 6 \sin \beta$$

$$\frac{9\sqrt{3}}{12} = \frac{\sin^3 \beta}{\cos \beta}$$

$$\frac{3\sqrt{3}}{4} = \frac{\sin^3 \beta}{\cos \beta} \Rightarrow \beta = 60^\circ$$

$$6\sqrt{3} - \frac{9\sqrt{3}}{2} \times \frac{2}{\sqrt{3}} + 6 \times \frac{1}{2} = \frac{3}{2} v_{\max}^2$$

$$6\sqrt{3} - 6 = \frac{3}{2} v_{\max}^2$$

$$4(\sqrt{3} - 1) = v_{\max}^2$$

$$v_{\max} = 2(\sqrt{3} - 1)^{\frac{1}{2}}$$

8. Ans (B,C)

$$\text{Current in the circuit } i = \frac{10}{R + \frac{R}{2}} = \frac{20}{3R}$$

$$\Rightarrow \text{Potential drop across PS} = iR = \frac{20}{3} V$$

$$\text{Potential drop across SQ} = i \left(\frac{R}{2} \right) = \frac{10}{3} V$$

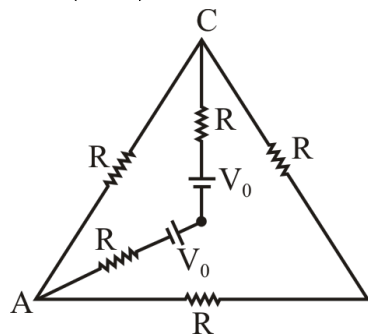
$$\Rightarrow \text{Potential gradient for PS} = \frac{20/3}{5} = \frac{4}{3} V/m$$

$$\text{Potential gradient for SQ} = \frac{10/3}{5} = \frac{2}{3} V/m$$

$$\Rightarrow \text{Length required for 5V} = \frac{5}{4/3} V = 3.75 \text{ m}$$

$$\text{Length required for 8V} = 5 + \frac{8 - \frac{20}{3}}{2/3} = 7 \text{ m}$$

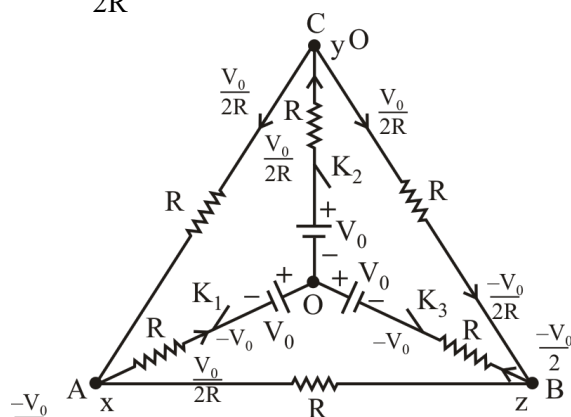
9. Ans (B,C)



$$E_{eq} = \frac{\frac{2V_0}{2R} + \frac{0}{2R}}{\frac{1}{2R} + \frac{1}{2R}} = \frac{\frac{V_0}{R}}{\frac{1}{R}} = V_0$$

$$R_{eq} = R$$

$$i_{AC} = \frac{V_0}{2R}$$



$$\begin{aligned} \frac{-\frac{V_0}{2R}}{R} + \frac{y - V_0}{R} + \frac{y - z}{R} &= 0 \Rightarrow 3y - x - z - V_0 = 0 \\ \frac{x - y}{R} + \frac{x + V_0}{R} + \frac{x - z}{R} &= 0 \Rightarrow 3x - y - z + V_0 = 0 \\ \frac{z - x}{R} + \frac{z + V_0}{R} + \frac{z - y}{R} &= 0 \Rightarrow 3z - x - y + V_0 = 0 \end{aligned}$$

$$z = \frac{x + y - V_0}{3}$$

$$3y - x - \left(\frac{x + y - V_0}{3} \right) - V_0 = 0$$

$$\frac{8}{3}y - \frac{4x}{3} - \frac{2V_0}{3} = 0$$

$$4y - 2x = V_0$$

$$3y - x - z - V_0 = 0$$

$$-3x + y + z - V_0 = 0$$

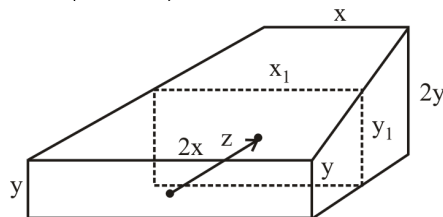
$$4y - 4x - 2V_0 = 0$$

$$2y - 2x = V_0$$

$$y = 0, x = -\frac{V_0}{2}$$

$$z = \frac{-\frac{V_0}{2} + 0 - V_0}{3} = \frac{-V_0}{2}$$

10. Ans (A,C,D)



$$A = x_1 y_1$$

$$x_1 = 2x - \frac{x}{L} \cdot z$$

$$y_1 = y + \frac{y}{L} \cdot z$$

$$dp = \frac{i^2 \rho dz}{x_1 y_1} \Rightarrow \frac{dp}{dz} = \frac{i^2 \rho}{\left(2x - \frac{x}{L}z\right) \left(y + \frac{y}{L}z\right)}$$

$$\frac{dp}{dz} = \frac{i^2 \rho}{xy \left(2 - \frac{z}{L}\right) \left(1 + \frac{z}{L}\right)} = \frac{i^2 \rho}{xy \left(2 - \frac{z^2}{L^2} + \frac{z}{L}\right)}$$

$$y = z - \frac{z^2}{L^2} + \frac{z}{L} \Rightarrow \frac{dy}{dz} = 0 - \frac{2z}{L^2} + \frac{1}{L} = 0$$

$$z = \frac{L}{2}$$

at

$$z = \frac{L}{2} \Rightarrow \frac{dp}{dz} = \frac{i^2 \rho}{xy \left(2 - \frac{1}{L^2} \times \frac{L^2}{4} + \frac{1}{2}\right)} = \frac{i^2 \rho}{xy \left(2 + \frac{1}{4}\right)}$$

$$\text{at } z = 0 \Rightarrow \frac{dp}{dz} = \frac{i^2 \rho}{xy(2)}$$

$$\left. \frac{d\rho}{dz} \right|_{\text{at } z=0} > \left. \frac{d\rho}{dz} \right|_{\text{at } z=\frac{L}{2}}$$

$$v_d = \frac{i}{neA} \quad v_d|_{\text{at } z=0} > v_d|_{\text{at } z=\frac{L}{2}}$$

PART-1 : PHYSICS

SECTION-III

4. Ans (5)

By Kepler's law, $T^2 \propto R^3$

5. Ans (2)

$$C_{AB} = 8\pi\epsilon_0 R \quad C_{BC} = 48\pi\epsilon_0 R \quad C_C = 16\pi\epsilon_0 R$$

$$\frac{1}{C_1} = \frac{1}{C_{AB}} + \frac{1}{C_{BC}} + \frac{1}{C_C} \Rightarrow C_1 = \frac{24\pi\epsilon_0 R}{5}$$

$$\frac{1}{C_2} = \frac{1}{C_{BC}} + \frac{1}{C_C} \Rightarrow C_2 = 12\pi\epsilon_0 R$$

$$\frac{1}{C_3} = \frac{1}{C_{AB}} + \frac{1}{C_C} \Rightarrow C_3 = \frac{16}{3}\pi\epsilon_0 R$$

$$C_4 = C_C = 16\pi\epsilon_0 R$$

$$(C_3 + C_4) - (C_1 + C_2) = \left(\frac{16}{3} + 4 - \frac{24}{5} \right) \pi\epsilon_0 R$$

$$= \frac{80 + 60 - 72}{15} = \frac{68}{15} \pi\epsilon_0 R$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A)

$$\text{For first cell } x = 0 - \frac{0.059}{1} \log \frac{[\text{Ag}^+]_{\text{anode}}}{[\text{Ag}^+]_{\text{cathode}}}$$

Let a mole/L be the solubility of Ag_2CrO_4

$$K_{sp} = (2a)^2 a$$

$$= 4a^3$$

$$\text{then } [\text{Ag}^+] = 2a$$

$$= 2 \left(\frac{K_{sp}}{4} \right)^{1/3}$$

$$X = 0 - \frac{0.059}{1} \log \frac{2 \times \left(\frac{p}{4} \right)^{1/3}}{0.1}$$

similarly for 2nd cell reaction

$$x = 0 - \frac{0.059}{1} \log \frac{q^{1/2}}{0.1}$$

for 3rd cell reaction

$$x = 0 - \frac{0.059}{1} \log \frac{1}{2 \times \left(\frac{s}{4} \right)^{1/3}}$$

At equilibrium $E_{\text{cell}} = x = 0$, so

$$\frac{2 \left(\frac{p}{4} \right)^{1/3}}{0.1} = \frac{q^{1/2}}{0.1} = \frac{3 \left(\frac{r}{27} \right)^{1/4}}{0.1} = \frac{1}{2 \left(\frac{s}{4} \right)^{1/3}} = 1$$

$$p = 5 \times 10^{-4}$$

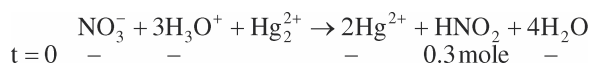
$$q = 10^{-2}$$

$$r = \frac{10^{-4}}{3}$$

$$s = 0.5$$

$$s > q > p > r$$

2. Ans (A)



Total charge $0.6 \text{ F} = 0.6 \times 96500 \text{ C}$

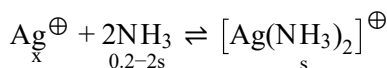
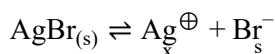
$$Q = It = 10 \times t = 0.6 \times 96500$$

$$t = \frac{0.6 \times 96500}{10} \text{ sec.} = 1.6 \text{ hour}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (A,B)



$$x \times s = 2 \times 10^{-10}$$

$$x \times s \frac{s}{x(0.2 - 2s)^2} = 16 \times 10^{-4}$$

$$\frac{s}{0.2 - 2s} = 4 \times 10^{-2}$$

$$\Rightarrow s = 8 \times 10^{-3} - 8 \times 10^{-2} s$$

$$\Rightarrow s = 8 \times 10^{-3}$$

$$x = \frac{10^{-7}}{4}$$

$$\text{Now, } \lambda_1 = \frac{K \times 1}{1000 \times x}$$

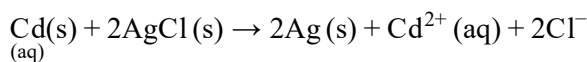
$$K_{\text{Ag}^+} = \lambda \times 1000 \times x$$

$$= 6 \times \frac{10^{-7}}{4}$$

$$= 1.5 \times 10^{-7} \text{ sm}^{-1}$$

6. Ans (A,C)

Net cell reaction is



$$\Delta G_{50^\circ\text{C}}^\circ = -nFE^\circ = -2 \times 96500 \times 0.6 = -115800 \text{ J}$$

$$\Delta G_{0^\circ\text{C}}^\circ = -nFE^\circ = -2 \times 96500 \times 0.7 = -135100 \text{ J}$$

$$\Delta S^\circ = nF \cdot \left(\frac{E_2^\circ - E_1^\circ}{T_2 - T_1} \right) = 2 \times 96500 \times \frac{0.6 - 0.7}{50}$$

$$= -386 \text{ J/K}$$

$$\Delta H^\circ = \Delta G^\circ + T \cdot \Delta S^\circ$$

$$= (-135100) + 273 \times (-386) = 240478 \text{ J}$$

7. Ans (A,C,D)

$$(A) E_a = \frac{k_1 E_{a1} + k_2 E_{a2}}{k_1 + k_2}$$

$$= \frac{10^{-3} \times 80 + 10^{-2} \times 100}{10^{-3} + 10^{-2}} = 98.18 \text{ kcal/mole}$$

$$(B) A = \frac{k}{e^{-E_a/RT}}$$

$$(D) f = e^{-E_a/RT}; E_{A1} \neq E_{A2}; f_1 \neq f_2$$

8. Ans (A,D)

2nd step is slowest

$$R = K_2[\text{CHCl}_3][\text{Cl}]$$

$$\frac{K_1}{K_{-1}} = \frac{[\text{Cl}]^2}{[\text{Cl}_2]} \Rightarrow [\text{Cl}] = \frac{K_1^{1/2}}{K_{-1}^{1/2}} [\text{Cl}_2]^{1/2}$$

$$R = \frac{K_2 K_1^{1/2}}{K_{-1}^{1/2}} [\text{CHCl}_3][\text{Cl}_2]^{1/2}$$

9. Ans (B,D)

(A) As per lock & key model, cavities on enzyme & structure of reactants are complimentary in shapes.

(B) Enzyme catalyst function only at optimum temperature.

(C) Vitamins are co-enzymes & Metal ions activations.

(D) Enzyme catalysed reactions are two step.

∴ Not elementary.

10. Ans (A,B,C)

Coagulative value



Coagulative power



Colloid is (-ve)

PART-2 : CHEMISTRY

SECTION-III

1. Ans (7)

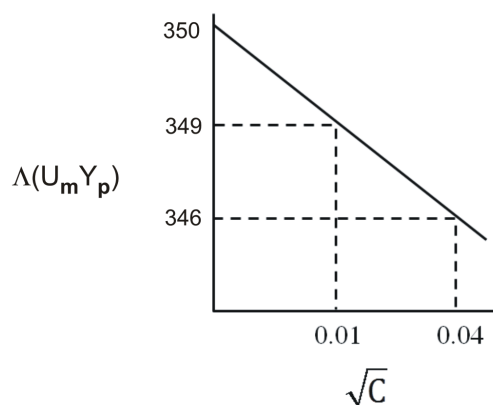
$$\Lambda^\circ(\text{Z}_m\text{X}_n) = m \times \lambda_{\text{Z}^{n+}}^\circ + n \times \lambda_{\text{X}^{m-}}^\circ = 250$$

$$5m + 8n = 25 \quad \dots\dots(1)$$

$$\Lambda^\circ(\text{V}_m\text{X}_n) = m \times \lambda_{\text{V}^{n+}}^\circ + n \times \lambda_{\text{X}^{m-}}^\circ = 440$$

$$100m + 80n = 440$$

$$5m + 4n = 22 \quad \dots\dots(2)$$



From the extrapolation of curve

$$\Lambda^\circ(\text{U}_m\text{Y}_p) = 350$$

$$m \times \lambda_{\text{U}^{p+}}^\circ + p \times \lambda_{\text{Y}^{m-}}^\circ = 350$$

$$25m + 100p = 350$$

$$m + 4p = 14 \quad \dots\dots(3)$$

Solving we get $(m + n + p) = 6.85$

2. Ans (3)

$$R = K(\text{NO})^a(\text{H}_2)^b$$

$$\frac{1.5}{0.25} = \left(\frac{372}{152}\right)^a \left(\frac{P}{P}\right)^b$$

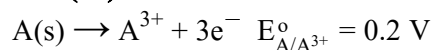
$$a = 2$$

$$\frac{1.6}{0.79} = \left(\frac{P}{P}\right)^a \left(\frac{289}{144}\right)^b$$

$$b = 1$$

$$(a + b) = 3$$

3. **Ans (5)**



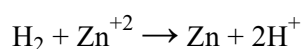
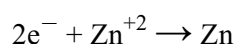
$$E_{\text{Net}}^\circ = 0.1 \text{ V}$$

$$E_{\text{cell}} = E_{\text{cell}}^\circ - 0.06/3 \log k$$

$$5 = \log (K_{\text{eq}})$$

$$K_{\text{eq}} = 10^5$$

4. **Ans (4)**



$$-0.6769 = -0.736 - \frac{0.0591}{2} \log \frac{[H^+]^2}{0.04}$$

$$0.0591 = -0.0591 \log \frac{[H^+]}{[0.2]}$$

$$\frac{[H^+]}{[0.2]} = \frac{1}{10}$$

$$[H^+] = 0.02$$

$$K = \frac{(0.02) \times (6.44 \times 10^{-3})}{0.4}$$

$$K = 3.22 \times 10^{-4}$$

$$X = 4$$

5. **Ans (9)**

$$K' = 1.8 \times 10^{-3} \times 55.55 \times 0.1$$

$$1 = \frac{2.303}{230.3} \log \frac{100 - 10}{100 - V_t} \Rightarrow V_t = 91 \text{ ml}$$

6. **Ans (6)**

Negatives solution will be precipitated by cation.

Eosin, Congo red, Ag Sol, Sb₂S₃ Sol, Clay sol.,

Charcoal sol.

7. **Ans (3)**

For nth order reaction :

$$r = k[A]^n$$

$$t_{1/2} = \frac{(2^{n-1} - 1)}{(n-1)ka_0^{n-1}} \quad (n \neq 1)$$

$$\frac{10^{-2}}{\frac{10^{-2}}{2}} = \left(\frac{20}{10}\right)^{n-1}$$

$$n = 2$$

$$R = k[A]^n = k[10^{-2}]^2$$

8. **Ans (5)**

Number of moles adsorbed

$$= \frac{100}{1000} \times 0.5 - \frac{100}{1000} \times 0.4 = 0.01$$

Number of molecules adsorbed

$$= 0.01 \times 6 \times 10^{23} = 6 \times 10^{21}$$

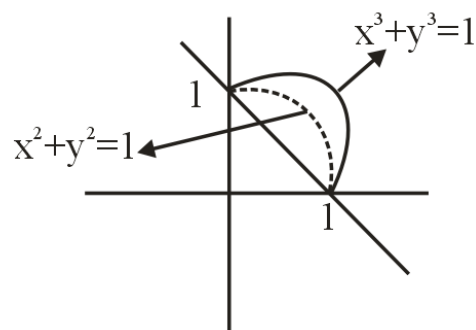
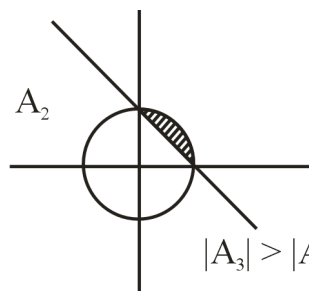
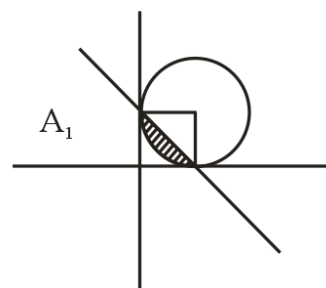
Surface area occupied

$$= \frac{3.01 \times 10^2}{6.02 \times 10^{21}} = \frac{1}{2} \times 10^{-19}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**



2. Ans (D)

$$\Rightarrow \frac{d(x^{3/2}) + d(y^{3/2})}{d(x^{3/2}) - d(y^{3/2})} = \frac{y^{3/2}}{x^{3/2}}$$

$$\Rightarrow \frac{du + dv}{du - dv} = \frac{v}{u}$$

Where $u = x^{3/2}$ and $v = y^{3/2}$

$$\Rightarrow udu + udv = vdu - vdv$$

$$\Rightarrow udu + vdv = vdu - udv$$

$$\Rightarrow \frac{udu + vdv}{u^2 + v^2} = \frac{vdu - udv}{u^2 + v^2}$$

$$\Rightarrow \frac{d(u^2 + v^2)}{u^2 + v^2} = -2d \tan^{-1} \left(\frac{v}{u} \right)$$

On integrating we get

$$\log(u^2 + v^2) = -2 \tan^{-1} \left(\frac{v}{u} \right) - 2c$$

$$\Rightarrow \frac{1}{2} \log(x^3 + y^3) + \tan^{-1} \left(\frac{y}{x} \right)^{\frac{3}{2}} + c = 0$$

3. Ans (C)

In I_2 , put $x + 1 = t$, then

$$I_2 = \int_{-2}^2 \frac{2t^2 + 11t + 14}{t^4 + 2} dt$$

$$= \int_{-2}^2 \frac{2x^2 + 11x + 14}{x^4 + 2} dx$$

$$\therefore I_1 + I_2 =$$

$$= \int_{-2}^2 \frac{x^6 + 3x^5 + 7x^4 + 2x^2 + 11x + 14}{x^4 + 2} dx$$

$$= \int_{-2}^2 \frac{(x^2 + 3x + 7)(x^4 + 2) + 5x}{x^4 + 2} dx$$

$$= \int_{-2}^2 (x^2 + 3x + 7) dx + 5 \int_{-2}^2 \frac{x}{x^4 + 2} dx$$

$$= 2 \int_0^2 (x^2 + 7) dx = \frac{100}{3}$$

The other integrals are zero, being integrals of

odd functions.

4. Ans (A)

$$\int \frac{(x^2 + 1) \tan^{-1} x + x \ln x \ln(\ln x)}{x(x^2 + 1) \ln x} dx$$

$$= \int \left(\tan^{-1} x \cdot \frac{1}{x \ln x} + \ln(\ln x) \cdot \frac{1}{1 + x^2} \right) dx$$

$$= \ln(\ln x) \cdot \tan^{-1} x + c$$

$$\phi(e) = 0 \Rightarrow c = 0$$

$$\therefore \text{G.E} : \lim_{x \rightarrow 1} \frac{\ln(\ln x) \cdot \tan^{-1} x}{\tan\left(\frac{\pi x}{2}\right)}$$

$$= \frac{\pi}{4} \cdot \lim_{x \rightarrow 1} \frac{\ln(\ln x)}{\tan\left(\frac{\pi x}{2}\right)} \left(\frac{\infty}{\infty} \right)$$

$$= \frac{\pi}{4} \lim_{x \rightarrow 1} \frac{\frac{1}{x \ln x}}{\sec^2\left(\frac{\pi x}{2}\right) \cdot \frac{\pi}{2}} = \frac{1}{2} \lim_{x \rightarrow 1} \frac{\cos^2\left(\frac{\pi x}{2}\right)}{\ln x}$$

$$\left(\frac{0}{0} \right)$$

$$= \frac{1}{2} \cdot \lim_{x \rightarrow 1} \frac{-\sin(\pi x) \cdot \frac{\pi}{2}}{\frac{1}{x}} = \frac{1}{2} (0) = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (B,C)

$$\text{Let } t = \frac{a-x}{1-ax} \Rightarrow x = \frac{t-a}{at-1} \Rightarrow dx = \frac{a^2-1}{(at-1)^2} dt$$

$$\therefore 1-x^2 = 1 - \left(\frac{t-a}{at-1} \right)^2 = \frac{(a^2-1)(t^2-1)}{(at-1)^2}$$

$$\therefore \int_0^a \frac{1-t^6}{t^2-1} dt = \int_0^a \frac{t^6-1}{t^2-1} dt = \frac{a^5}{5} + \frac{a^3}{3} + a = F(a)$$

$$F(1) = \frac{1}{5} + \frac{1}{3} + 1 = \frac{3+5+\sqrt{15}}{15} = \frac{23}{15}$$

6. Ans (A,B,D)

$$\text{Let } y = \sin x \text{ and } \frac{dy}{dx} = \cos x$$

$$\Rightarrow \frac{dy + dx}{dx - dy} = \frac{1 + \cos x}{1 - \cos x} = \cot^2 \left(\frac{x}{2} \right)$$

$$\Rightarrow \tan^2 \left(\frac{x}{2} \right) d(x+y) = d(x-y)$$

$$\Rightarrow \int (x-y) d(x-y) = \frac{(x-y)^2}{2} + c$$

$$\text{Also } \frac{d^2x}{dy^2} = \frac{\frac{-d^2y}{dx^2}}{\left(\frac{dy}{dx} \right)^3} = \frac{-(\sin x)}{\cos^3 x} = \frac{\sin x}{\cos^3 x}$$

7. Ans (B,D)

Area bounded by $f(x) = |\cos x|$, $g(x) = \sin x$

$$\int_0^{\pi} |f(x) - g(x)| dx = 4 \int_0^{\pi/4} (\cos x - \sin x) dx$$

$$= 4(\sqrt{2} - 1)$$

$$\int_{\pi}^{2\pi} |f(x) - g(x)| dx = \int_{\pi}^{2\pi} (|\cos x| - \sin x) dx = 4$$

So if x is even $A = 2x\sqrt{2}$

N is odd $A = 2(n-1)\sqrt{2} + 4(\sqrt{2} - 1)$

$$= 2(n+1)\sqrt{2} - 4$$

8. Ans (A,B,C)

(A, B, C) for $x \in (0, \frac{\pi}{2})$

$$0 < \sin x < 1 < \operatorname{cosec} x < \infty$$

$$\text{So, } \sin x^{\operatorname{cosec} x} < \sin x^{\sin x} < \operatorname{cosec} x^{\sin x} < \operatorname{cosec} x^{\operatorname{cosec} x}$$

9. Ans (A,B,C)

$$(2n+1)[I_{n+1} - I_n] = (2n+1)$$

$$\int \frac{-\cos^{2n} x (1 - \cos^2 x)}{\sin x} dx = \cos^{2n+1} x$$

$$\therefore \phi(x) = \cos^{2n+1} x \Rightarrow \phi'(x)$$

$$= (2n+1)\cos^{2n} x (-\sin x) = f(n, x)$$

10. Ans (A,C)

(A) Use $\tan^{-1} x < x$

$$(C) \text{ LHS} = \left| \int_a^b (f(x) - f(a)) dx \right| \leq$$

$$\int_a^b |f(x) - f(a)| dx \leq \int_a^b |x - a| dx \leq \int_a^b (x - a) dx$$

$$\leq \frac{(b-a)^2}{2}$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (4)

$$\int_0^{65\frac{\pi}{4}} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})}$$

$$= \int_0^{16\pi+\frac{\pi}{4}} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})}$$

$$\left(\because \int_a^{a+T} f(x) dx = \int_0^T f(x) dx \right)$$

$$= \int_0^{16\pi} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})}$$

$$= \int_0^{8(2\pi)} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})}$$

$$= 8 \int_0^{2\pi} \frac{dx}{(1+2^{\cos x})(1+2^{\sin x})} \quad \dots(1)$$

$$\left(\because \int_a^{nT} f(x) dx = n \int_0^T f(x) dx \right)$$

$$= 8 \int_0^{2\pi} \frac{dx}{(1+2^{\cos x})(1+2^{-\sin x})} \quad \dots(2)$$

$$\left(\because \int_0^a f(x) dx = \int_0^a f(a-x) dx \right)$$

(1) + (2)

$$I = 8 \int_0^{2\pi} \frac{(1+2^{\sin x})}{(1+2^{\cos x})(1+2^{\sin x})} dx$$

$$I = 4 \int_0^{2\pi} \frac{1}{(1+2^{\cos x})} dx$$

$$\left(\because \int_a^{nT} f(x) dx = n \int_0^T f(x) dx \right)$$

$$I = 4 \times 2 \int_0^{\pi} \frac{1}{(1+2^{\cos x})} dx \quad \dots(3)$$

$$I = 8 \int_0^{\pi} \frac{1}{(1+2^{-\cos x})} dx \quad \dots(4)$$

$$2I = 8 \int_0^{\pi} 1 dx \quad \dots(5)$$

$$I = 4\pi$$

2. Ans (7)

$$S_k(x) = \left\{ 1 - \int_0^1 S_{k-1}(t) dt \right\} x + k \int_0^x S_{k-1}(t) dt$$

$$S_k(0) = 0$$

$$S_k(x) = x + k \int_0^x S_{k-1}(t) dt - \int_0^1 S_{k-1}(t) dt$$

$$S'_k(x) = 1 + k S_{k-1}(x) - 0$$

$$(k=1) S'_1(x) = 1 + S_0(x) = 1 + x$$

$$S_1(x) = \frac{x^2}{2} + x + C$$

$$\therefore S_1(0) = 0 \therefore S_1(x) = \frac{x^2}{2} + x$$

$$(k=2) S'_2(x) = 1 + 2S_1(x)$$

$$= 1 + x^2 + 2x = (x+1)^2$$

$$\therefore S_2(x) = \frac{(x+1)^3}{3} + C \therefore S_2(0) = 0$$

$$\therefore S_2(x) = \frac{(x+1)^3 - 1}{3}$$

$$\therefore S_2(3) = \frac{4^3 - 1}{3} = 21$$

$$(k=3) S'_3(x) = 1 + 3S_2(x)$$

$$= 1 + (x+1)^3 - 1 = (x+1)^3$$

$$\therefore S_3(x) = \frac{(x+1)^4}{4} + C \therefore S_3(0) = 0 \therefore C = -\frac{1}{4}$$

$$\therefore S_3(x) = \frac{(x+1)^4 - 1}{4}$$

$$C_1 = 1 - \int_0^1 x dx = \frac{1}{2}$$

$$C_2 = 1 - \int_0^1 \left(\frac{x^2}{2} + x \right) dx = 1 - \left(\frac{1}{6} + \frac{1}{2} \right) = \frac{1}{3}$$

$$C_3 = 1 - \int_0^1 \left(\frac{x^3}{3} + x^2 + x \right) dx = 1 - \left(\frac{1}{12} + \frac{1}{3} + \frac{1}{2} \right) = \frac{1}{12}$$

$$C_3 = 1 - \int_0^1 S_2(x) dx = 1 - \int_0^1 \frac{(x+1)^3 - 1}{3} dx$$

$$= 1 - \left[\frac{(x+1)^4}{12} - \frac{x}{3} \right]_0^1 = 1 - \left[\left(\frac{16}{12} - \frac{1}{3} \right) - \frac{1}{12} \right]$$

$$= 1 - \left(1 - \frac{1}{12} \right) = \frac{1}{12}$$

$$\therefore S_2(3) + 6C_3 = 21 + 6 \times \frac{1}{12} = 21.5$$

3. Ans (0)

$$\text{If } \int \frac{3x^2 + 2x}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5} dx$$

$$\int \frac{3x^2 + 2x}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5} dx$$

$$\int \frac{3x^2 + 2x}{(x^3 + x^2 + 1)^2 + 4} dx = \frac{1}{2} \tan^{-1} \left(\frac{x^3 + x^2 + 1}{2} \right) + C$$

$$F(x) = \frac{1}{2} \tan^{-1} \left(\frac{x^3 + x^2 + 1}{2} \right) + C$$

$$\therefore F(1) - F(0) = \frac{1}{2} \left(\tan^{-1} \frac{3}{2} - \tan^{-1} \frac{1}{2} \right) = \frac{1}{2} \tan^{-1} \frac{4}{7}$$

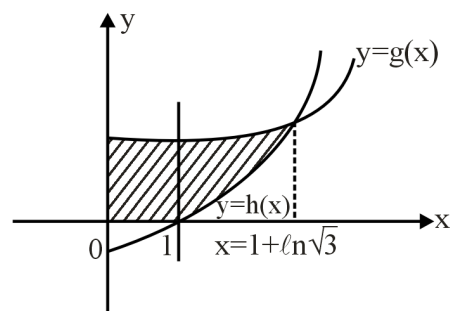
$$\therefore 0 < F(1) - F(0) < \frac{1}{2} \cdot \frac{\pi}{2} < 1$$

$$\therefore [F(1) - F(0)] = 0$$

4. Ans (3)

$$\text{Here } f(x) = \begin{cases} 0 & x \leq 1 \\ e^{x+1} - e^{1-x} & x \geq 1 \end{cases}$$

$$\& g(x) = \frac{1}{2} (e^{x-1} + e^{1-x})$$



$$\text{Solve } f(x) \& g(x) \Rightarrow x = 1 + \ln \sqrt{3}$$

$$\text{So bounded area} = \int_1^{1+\ln \sqrt{3}} \frac{1}{2} (e^{x-1} + e^{1-x}) - (e^{x-1} - e^{1-x}) dx$$

$$= \left[-\frac{1}{2} e^{x-1} - \frac{3}{2} e^{1-x} \right]_1^{1+\ln \sqrt{3}}$$

$$= \left[\left(-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right) + 2 \right] = 2 - \sqrt{3}$$

5. Ans (9)

$$y^2 = 4\lambda x, P(\lambda, 2\lambda)$$

Slope of the tangent to the parabola at point P

$$\frac{dy}{dx} = \frac{4\lambda}{2y} = \frac{4\lambda}{2 \times 2\lambda} = 1$$

Slope of the tangent to the ellipse at P

$$\frac{2x}{a^2} + \frac{2yy'}{b^2} = 0$$

As tangents are perpendicular $y' = -1$

$$\Rightarrow \frac{2\lambda}{a^2} - \frac{4\lambda}{b^2} = 0 \Rightarrow \frac{a^2}{b^2} = \frac{1}{2}$$

$$\Rightarrow e = \sqrt{1 - \frac{1}{2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{a}{b} = \pm \frac{1}{\sqrt{2}}; e = \frac{1}{\sqrt{2}}$$

$$15 \left(\frac{1}{2} \right) + 3 \left(\frac{1}{2} \right) = K$$

$$\frac{15}{2} + \frac{3}{2} = K \Rightarrow K = 9$$

6. Ans (1)

$$(bx + cy + a)dy = (ax - by + a)dx$$

$$\Rightarrow b d(xy) + c y dy + a dy - ax dx - a dx = 0$$

Integrating on both sides

$$b(xy) + \frac{cy^2}{2} + ay - \frac{ax^2}{2} - ax = k \quad \dots(1)$$

Equations (1) represents a circle then $\frac{a}{2} = -\frac{c}{2}$

$$\Rightarrow a = -c, b = 0$$

$$\frac{a}{2}x^2 + \frac{a}{2}y^2 + ax - ay + k = 0 \Rightarrow x^2 + y^2 + 2x$$

$$- 2y + \frac{2k}{a} = 0$$

$$\text{Centre } (\alpha, \beta) = (-1, 1), \alpha + 2\beta = -1 + 2 = 1$$

7. Ans (5)

Clearly

$$f'(x) = -3x^{-4} \left(\frac{x^{12}}{1+x^{12}} + x^4 \right) + 3 \left(\frac{x^8}{1+x^{12}} \right)$$

$$\Rightarrow f'(x) = -3 \quad \forall x$$

$$\Rightarrow f(x) = -3x + c \quad (\text{But } f(1) = 0)$$

$$\therefore c = 3$$

$$\therefore f(x) = -3x + 3 \Rightarrow y = -3x + 3 \Rightarrow 3x + y - 3 = 0$$

$$\text{Area} = \frac{(-3)^2}{2(3)(1)} = \frac{3}{2} \left\{ \begin{array}{l} p = 3 \\ q = 2 \end{array} \right. \Rightarrow p + q = 5$$

8. Ans (4)

$$v = \int_0^{\pi/4} (1 + \tan x)^2 dx$$

$$= \int_0^{\pi/4} \left(1 + \tan \left(\frac{\pi}{4} - x \right) \right)^2 dx$$

$$= \int_0^{\pi/4} \left(1 + \frac{1 - \tan x}{1 + \tan x} \right)^2 dx$$

$$= 4 \int_0^{\pi/4} \left(\frac{1}{1 + \tan x} \right)^2 dx$$

$$= 4 \int_0^{\pi/4} \left(\frac{\cos x}{\cos x + \sin x} \right)^2 dx = 4u$$

$$\therefore \frac{v}{u} = 4$$